CLOSED-FORM SYMPLECTIC PACKINGS OF THE FOUR-BALL BY EQUAL BALLS

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ABSTRACT. For every integer $k \geq 10$ and every $\varepsilon \in (0,1)$ we construct explicit C^4 symplectic embeddings $f_0, \ldots, f_{k-1} \colon \mathbb{B}^\circ(r_+) \to \mathbb{B}^\circ(R)$ in $\mathbb{R}^4$, where $r_+ > 1$ and $R = (k/(1-\varepsilon))^{1/4}$, whose images are pairwise disjoint, together with an explicit non-membership-certifying left inverse on the target ball. Unlike the Hamiltonian flow construction of Paper I, no ordinary differential equation is integrated anywhere: each f_i is a finite composition of elementary closed-form maps together with scalar roots of strictly monotone one-variable equations (one Kepler equation in the inverse strip map and generating-function angle/radial equations in two planar keyholes). The geometry adapts the degree-9 disc-bundle mechanism of Paper I—linear comparison slope $3\sqrt{k}-9$ and three marked fibres mapped to the line at infinity of $\mathbb{CP}^2$—into direct Darboux charts, generating-function keyholes with explicit full-disc nesting and slit bounds, C^4 deck-equivariant toric twists, and a piecewise closed-form transfer across the three edges of the moment simplex of $\mathbb{CP}^2$. All parameter inequalities hold analytically for every $k \geq 10$ and $\varepsilon \in (0,1)$. A normalised Liouville primitive gives an explicit generator formula for the radial–Alexander collar stage, together with abstract quantitative ODE trajectory certificates. Sections 1–7 prove the exact-real algorithm implemented by `program/solution_embedding.py`, and finite-precision evaluation is checked numerically; lifting these direct maps to a standalone global Hamiltonian flow (distinct from the C_c^∞ flow already proved in Paper I) is left to future work.

1. Introduction and main theorem

1.1. **Statement of the result.** Write $\mathbb{R}^4$ with coordinates $X = (q_1, q_2, p_1, p_2)$ and standard symplectic form $\omega_0 = dq_1 \wedge dp_1 + dq_2 \wedge dp_2$. Let $\mathbb{B}(r) = \{X \in \mathbb{R}^4 : |X| \leq r\}$ and $\mathbb{B}^\circ(r) = \{X \in \mathbb{R}^4 : |X| < r\}$ denote the closed and open Euclidean balls of radius r about the origin; in capacity units, $B(c) = \{\pi|z|^2 \leq c\}$ and $B^\circ(c) = \{\pi|z|^2 < c\}$, so $\mathbb{B}(1) = B(\pi)$ and $\mathbb{B}(R) = B(\pi R^2)$.

Paper I [Son26c] constructs, for every $k \geq 10$ and $\varepsilon \in (0,1)$, a C^∞ compactly supported Hamiltonian on $\mathbb{R} \times \mathbb{R}^4$ whose time-one flow packs k closed unit balls into $\mathbb{B}^\circ((k/(1-\varepsilon/4))^{1/4})$, verified in Lean 4. Several steps of that construction—the handle Moser isotopy and the deformation-to-the-normal-cone transport and return—are defined as time-one maps of stiff vector fields. When only direct evaluation of the packing embeddings and their inverses is required, every differential equation can be replaced by closed-form charts and generating functions.

Relation to prior work. Biran proved full packings of the four-ball by every $k \geq 9$ equal balls [Bir97]; explicit constructions include those of Traynor and the methods surveyed by Schlenk [Tra95, Sch18]. The present maps develop Biran's cubic disc-bundle picture [Bir01] and the twisted disc-bundle mechanism introduced in higher dimensions by OpenAI [Ope26]. Paper I supplies the new four-dimensional comparison identity and return mechanism; here they are realised by direct charts and scalar monotone roots, eliminating ODE integration from map evaluation. The strip map uses the triangular rearrangement of Rosenblatt and Knothe [Ros52, Kno57]. These direct C^4 maps complement the smooth Hamiltonians of Paper I

Version 1.2.0; companion to [Son26c, Son26b, Son26a] in `https://github.com/s-JoL/Symplectic-Ball-Packing` (all versions: doi:10.5281/zenodo.23756575).

in relation to Epoch's problem [Epo26]: they do not by themselves supply its Hamiltonian ODE interface.

Theorem 1.1 (Closed-form C^4 equal-ball packings)**.** *Let $k \geq 10$ be an integer and $\varepsilon \in (0,1)$. Put $\varepsilon_c = \min(\varepsilon, \frac{1}{100})$, $\eta = \varepsilon_c/(1+\sqrt{1-\varepsilon_c}) = 1-\sqrt{1-\varepsilon_c}$, $m_s = \min(2\cdot 10^{-6}, \frac{1}{8}\eta)$, and*

$$R_{\mathrm{req}} = \left(\frac{k}{1-\varepsilon}\right)^{1/4}, \quad R_c = \left(\frac{k}{1-\varepsilon_c}\right)^{1/4}, \quad R_s = (1-m_s)R_c, \quad r_+ = (1-m_s)\left(\frac{1-\eta/2}{1-\eta}\right)^{1/2}.$$

Then $r_+ > 1$ and $R_s < R_c \leq R_{\mathrm{req}}$, and with the explicit parameter tuple $\Pi^*(k, \varepsilon_c; \frac{1}{4})$ of Definition 7.1, the compositions

$$f_i = S_{\mathrm{out}} \circ \chi_\kappa \circ \mathcal{W} \circ \mathcal{B} \circ \Pi_{\mathrm{post}} \circ \Phi_g \circ \Psi_S \circ F^{(i)} \circ C_i \circ \mathrm{KR} \circ S_{\mathrm{in}} \qquad (0 \leq i \leq k-1)$$

defined in Sections 2–7 satisfy over $\mathbb{R}$:

- *(i) Each $f_i \colon \mathbb{B}^\circ(r_+) \to \mathbb{R}^4$ is C^4, injective, and symplectic ($f_i^*\omega_0 = \omega_0$), mapping $\mathbb{B}^\circ(r_+)$ onto an open subset of $\mathbb{B}^\circ(R_s) \subset \mathbb{B}^\circ(R_{\mathrm{req}})$.*
- *(ii) The open images $f_0(\mathbb{B}^\circ(r_+)), \ldots, f_{k-1}(\mathbb{B}^\circ(r_+))$ are pairwise disjoint; in particular the restrictions $f_i|_{\mathbb{B}(1)}$ pack k closed unit balls disjointly into $\mathbb{B}^\circ(R_{\mathrm{req}})$ with occupied volume fraction $k/R_{\mathrm{req}}^4 = 1-\varepsilon$.*
- *(iii) Each stage factor of f_i has an explicit inverse, and the composite inverse procedure* Preimage(j, Y) *of Section 7 satisfies* Preimage$(i, f_i(X)) = (X, \mathrm{true})$ *for every $X \in \mathbb{B}^\circ(r_+)$ and $0 \leq i < k$, while certifying non-membership ($|X| > 1$ or* ok $=$ false*) whenever $Y \notin f_j(\mathbb{B}(1))$.*

1.2. **Architecture and correspondence with code.** Table 1 lists the stage maps of f_i in evaluation order together with their domains, symplectic forms, and modular source files in `program/embedding_src/` (bundled into `program/solution_embedding.py` by `program/build_embedding.py`). Regularity is C^ω on the algebraic/trigonometric charts, C^8 on the planar keyholes ($m = 4$), C^7 on the four-dimensional keyhole suspensions ($S_4' \in C^7$ enters the conjugate drift), and C^4 on the marked-fibre twists ($m = 2$ plateau bumps); hence $f_i \in C^4$.

Stage	Map	Domain → target & form	Section / file
1. Input scale	$S_{\mathrm{in}}(q_1, q_2, p_1, p_2)$	$\mathbb{B}^\circ(r_+) \to B^\circ(a)$, conformal $\pi^{-1}R_s^{-2}$	§7; `chain.py`
2. Strip map	$\mathrm{KR}(x, y, w)$	$B^\circ(a) \xrightarrow{\sim} \mathcal{C}_a = \{0 < X < a-p, 0 < T < 1\}$	§2.2; `keyhole.py`
3. Cell to flat	$C_i(X, T, w)$	$i < 9$: group windows; $i \geq 9$: fan stack	§4; `chain.py`
4–5. Comparator	$F^{(i)}$ ($F_2 \circ F_1$ for $i \geq 9$)	Keyhole $KS_{\tau(G)}$ + slit shift $c(G)$	§4; `comparator.py`
6–7. Twists	$\Psi = \Phi_g \circ \Psi_S$	Routes $\mathcal{M}_1, \mathcal{M}_2, \mathcal{M}_3$ into free corridor	§5; `twists.py`
8. Post chart	$\Pi_{\mathrm{post}}(G, v, w)$	Deck-adapted Darboux chart on $\{p < h_K\}$	§6; `chain.py`
9. Corner band	$\mathcal{B}$ (`Band`)	Corner keyhole K_B along $\sigma = j(1-3p)$	§6; `transfer.py`
10. Simplex glue	$\mathcal{W}(\sigma, \psi, p, \vartheta)$	4 sector pieces + 3 bands → $\mathbb{CP}^2 \setminus L_\kappa$	§6; `transfer.py`
11. Ball chart	$S_{\mathrm{out}} \circ \chi_\kappa([Z])$	$\mathbb{CP}^2 \setminus L_\kappa \xrightarrow{\sim} B^\circ(1) \to \mathbb{B}^\circ(R_s)$	§6; `transfer.py`

TABLE 1. Stage decomposition ('STAGES(i)' in `solution_embedding.py`). For a planar fibre coordinate $w = w_x + iw_y \in \mathbb{C}$, $p = \pi|w|^2$ and $\theta = \arg(w)/(2\pi) \in \mathbb{R}/\mathbb{Z}$ satisfy $\omega_V = dw_x \wedge dw_y = dp \wedge d\theta$.

2. Smooth primitives and the Knothe–Rosenblatt strip map

2.1. **Trigonometric steps, ramps, and plateau bumps.** For an integer $m \geq 1$, let $c_0 = 2^{-2m}\binom{2m}{m}$ and $a_j = 2(-1)^j\binom{2m}{m-j}/\binom{2m}{m}$ $(1 \leq j \leq m)$, so that $\sin^{2m}(\pi u) = c_0\big(1 + \sum_{j=1}^{m} a_j \cos(2\pi j u)\big)$. Define $S_m\colon \mathbb{R} \to [0,1]$ by $S_m(u) = 0$ for $u \leq 0$, $S_m(u) = 1$ for $u \geq 1$, and

$$S_m(u) = u + \sum_{j=1}^{m} \frac{a_j}{2\pi j} \sin(2\pi j u) \qquad (0 \leq u \leq 1).$$

Lemma 2.1 (Properties of S_m, I_m, R_m, and $\beta_{w_0,w_1,m}$)**.** *(i) $S_m'(u) = \sin^{2m}(\pi u)/c_0$ on $[0,1]$ vanishes to order $2m$ at $u = 0, 1$. Hence $S_m \in C^{2m}(\mathbb{R})$, $S_m' \in C^{2m-1}(\mathbb{R})$, S_m is strictly increasing on $[0,1]$ with $S_m(1-u) = 1 - S_m(u)$, convex on $[0, \frac{1}{2}]$ and concave on $[\frac{1}{2}, 1]$; in particular $S_m(u) \leq u$ for all $u \in [0, \frac{1}{2}]$.*

(ii) The antiderivative $I_m(u) = \int_0^{\mathrm{clamp}(u,0,1)} S_m(t)\,dt$ equals $\frac{1}{2}u^2 + \sum_{j=1}^{m} \frac{a_j}{(2\pi j)^2}(1 - \cos(2\pi j u))$ on $[0,1]$ with $I_m(1) = \frac{1}{2}$. The convex ramp $R_m(u) = I_m(\mathrm{clamp}(u,0,1)) + \max(u-1, 0)$ belongs to $C^{2m+1}(\mathbb{R})$, equals 0 on $(-\infty, 0]$ and $u - \frac{1}{2}$ on $[1, \infty)$, and satisfies $\max(0, u - \frac{1}{2}) \leq R_m(u) \leq \max(0, u)$ for all $u \in \mathbb{R}$.

(iii) For $0 < w_0 < w_1$, $\beta_{w_0,w_1,m}(x) = 1 - S_m\big((|x| - w_0)/(w_1 - w_0)\big)$ belongs to $C^{2m}(\mathbb{R})$ ($|x|$ is not differentiated on $(-w_0, w_0)$ where $\beta_{w_0,w_1,m} \equiv 1$, and all derivatives up to order $2m$ vanish at $|x| = w_0, w_1$), and its primitive $\mathcal{B}(x) = \int_{-\infty}^{x} \beta_{w_0,w_1,m}(t)\,dt \in C^{2m+1}(\mathbb{R})$ has total mass $\mathcal{B}(\infty) = w_0 + w_1$.

Proof. Since $\sin(\pi u)$ has simple zeros at $u = 0, 1$, $\sin^{2m}(\pi u)$ vanishes to order $2m$, giving $S_m \in C^{2m}(\mathbb{R})$. On $(0, \frac{1}{2})$, $S_m''(u) = (2m\pi/c_0)\sin^{2m-1}(\pi u)\cos(\pi u) > 0$, so convexity with $S_m(0) = 0$, $S_m(\frac{1}{2}) = \frac{1}{2}$ yields $S_m(u) \leq u$ on $[0, \frac{1}{2}]$ and concavity on $[\frac{1}{2}, 1]$ via $S_m(1-u) = 1 - S_m(u)$. Integrating the symmetry identity gives $I_m(1) = \frac{1}{2}$, and $R_m' = S_m \in C^{2m}(\mathbb{R})$ gives $R_m \in C^{2m+1}(\mathbb{R})$ with supporting lines 0 and $u - \frac{1}{2}$. Part (iii) follows from $w_0 > 0$ and $\int_{-\infty}^{\infty} \beta_{w_0,w_1,m} = 2w_0 + 2(w_1 - w_0)(1 - I_m(1)) = w_0 + w_1$. □

2.2. **The fibrewise Knothe–Rosenblatt strip map.** Let $B^\circ(a) = \{(x,y,w) \in \mathbb{R}^2 \times \mathbb{C} : \pi(x^2+y^2) + p < a\}$ $(p = \pi|w|^2)$ and $\mathcal{C}_a = \{(X,T,w) \in \mathbb{R}^2 \times \mathbb{C} : p < a,\ 0 < X < a - p,\ 0 < T < 1\}$.

Proposition 2.2 (Knothe–Rosenblatt ball-to-cell map)**.** *For $(x,y,w) \in B^\circ(a)$, set $r(p) = \sqrt{(a-p)/\pi} > 0$ and $c(x,p) = \sqrt{r(p)^2 - x^2} > 0$, and define*

$$\begin{aligned} X(x,p) &= r^2\Big(\frac{\pi}{2} + \arcsin\frac{x}{r} + \frac{xc}{r^2}\Big) = \int_{-r}^{x} 2\sqrt{r^2 - s^2}\,ds, \\ T(x,y,p) &= \frac{1}{2} + \frac{y}{2c}, \qquad C(x,y,p) = \frac{X - xc}{\pi r^2}\,\frac{y}{2c}, \qquad \mathrm{KR}(x,y,w) = \big(X, T, we^{2\pi i C}\big). \end{aligned} \tag{1}$$

Then $\mathrm{KR}\colon B^\circ(a) \to \mathcal{C}_a$ is a real-analytic symplectomorphism satisfying $\mathrm{KR}^*(dX \wedge dT + dw_x \wedge dw_y) = dx \wedge dy + dw_x \wedge dw_y$. Its inverse on $\mathcal{C}_a$ solves Kepler's equation $2\alpha + \sin(2\alpha) = 2(X/r^2 - \pi/2)$ for the unique $\alpha \in (-\frac{\pi}{2}, \frac{\pi}{2})$, sets $x = r\sin\alpha$, $c = r\cos\alpha$, $y = (2T-1)c$, C by (1), and returns $(x, y, we^{-2\pi i C})$.

Proof. For $(x,y,w) \in B^\circ(a)$ we have $p < a$, $|x| < r(p)$, and $|y| < c(x,p)$, so $r, c > 0$ and $|x/r| < 1$; hence KR is real-analytic on all of $B^\circ(a)$ (including $w = 0$). At fixed $p < a$, $X(x,p) \in (0, \pi r^2) = (0, a - p)$ has $\partial_x X = 2c > 0$, and substituting $x = r\sin\alpha$ gives $X/r^2 = \frac{\pi}{2} + \alpha + \frac{1}{2}\sin(2\alpha)$, whose derivative $1 + \cos(2\alpha) = 2\cos^2\alpha > 0$ on $(-\frac{\pi}{2}, \frac{\pi}{2})$ makes KR^{-1} the exact inverse on $\mathcal{C}_a$.

For symplecticity it suffices by real-analyticity to check $w \neq 0$. Put $w' = we^{2\pi i C}$, so $p' = p$ and $dw'_x \wedge dw'_y = dp \wedge d(\theta + C) = dp \wedge d\theta + dp \wedge d_{x,y}C$. Since $d_{x,y}X \wedge d_{x,y}T = (2c\,dx) \wedge (\frac{dy}{2c} + \partial_x T\,dx) = dx \wedge dy$, expanding $dX \wedge dT$ gives

$$dX \wedge dT + dw'_x \wedge dw'_y = dx \wedge dy + dp \wedge d\theta + dp \wedge \big(\partial_p X\, d_{x,y}T - \partial_p T\, d_{x,y}X + d_{x,y}C\big).$$

Differentiating $X = \int_{-r}^{x} 2\sqrt{r^2 - s^2}\,ds$ with respect to $r^2 = (a-p)/\pi$ yields $\partial_p X = -\frac{1}{\pi}\int_{-r}^{x}(r^2 - s^2)^{-1/2} ds = -\frac{X - xc}{\pi r^2}$, while $\partial_p c = -\frac{1}{2\pi c}$ gives $\partial_p T = \frac{y}{4\pi c^3}$. Since $C = (-\partial_p X)(T - \frac{1}{2})$ and $\partial_x\big(\frac{X-xc}{\pi r^2}\big) = \frac{2c - (c - x^2/c)}{\pi r^2} = \frac{1}{\pi c}$, we obtain

$$\partial_p X\, d_{x,y}T - \partial_p T\, d_{x,y}X + d_{x,y}C = -\frac{y}{4\pi c^3}(2c\,dx) - \frac{y}{2c}\Big(-\frac{dx}{\pi c}\Big) = 0. \qquad \square$$

3. The smooth planar keyhole and full-disc bounds

3.1. **Generating-function construction.** Let $r_a > 0$, $d > 0$, $0 < W_1 < W_2 \le \frac{1}{6}$, $0 < \rho_1 < 1$, $\beta > 0$, $X_w > 0$, and $r_h > 0$ satisfy

$$(W_1 + W_2)(1 + \beta - \rho_1) = \beta, \tag{2a}$$

$$0 < r_h \le d \sin(2\pi W_1), \tag{2b}$$

$$\rho_1 \pi (d + r_a)^2 + \frac{1}{2} X_w < \pi (d - r_h)^2. \tag{2c}$$

Notice that (2b), with $W_1 < W_2 \le 1/6$, implies $r_h < d$. Around the eccentric centre $c_* = (-d, 0) \in \mathbb{C}$, write $w = c_* + Re^{2\pi i\phi}$ with action $X = \pi R^2 \ge 0$ and angle $\phi \in [-\frac{1}{2}, \frac{1}{2}]$. Set $M(X) = X_w R_4(X/X_w)$, so $M'(X) = S_4(\min(X/X_w, 1)) \in [0,1]$ and $X - \frac{1}{2}X_w \le M(X) \le X$ for all $X \ge 0$ (Lemma 2.1(ii)). Define the odd map Ψ on $[-\frac{1}{2}, \frac{1}{2}]$ by $\Psi(0) = 0$ and

$$\Psi'(\psi) = \rho_1 + (1 + \beta - \rho_1) S_4\Big(\frac{|\psi| - W_1}{W_2 - W_1}\Big) \qquad \big(|\psi| \le \tfrac{1}{2}\big).$$

Lemma 3.1 (Properties of $\Psi(\psi) - \psi$)**.** *Under* (2a), $\Psi(\pm\frac{1}{2}) = \pm\frac{1}{2}$, *and* $\Psi(\psi) - \psi$ *extends* 1*-periodically to an odd function in* $C^9(\mathbb{R})$. *On* $[0, \frac{1}{2}]$, $\Psi(\psi) - \psi \le 0$, *with* $\Psi(\psi) - \psi = -\beta(\frac{1}{2} - \psi)$ *on* $[W_2, \frac{1}{2}]$, $0 \le \psi - \Psi(\psi) \le (1 - \rho_1)\psi < W_2$ *on* $[0, W_2]$, *and* $|\Psi(\psi) - \psi| \le \frac{1}{2}\beta$ *everywhere.*

Proof. Using $I_4(1) = \frac{1}{2}$, $\Psi(\frac{1}{2}) - \frac{1}{2} = (\rho_1 - 1)W_1 + \frac{1}{2}(\rho_1 + \beta - 1)(W_2 - W_1) + \beta(\frac{1}{2} - W_2) = \frac{1}{2}\big(\beta - (W_1 + W_2)(1 + \beta - \rho_1)\big) = 0$ by (2a). Since $\Psi' \equiv 1 + \beta$ near $\pm\frac{1}{2}$, $\Psi(\psi) - \psi$ is affine of slope β across $\pm\frac{1}{2}$ (mod 1), hence C^9 on $\mathbb{R}/\mathbb{Z}$. On $[0, \frac{1}{2}]$, Ψ' is non-decreasing, so $\Psi(\psi) - \psi$ is convex and vanishes at $0, \frac{1}{2}$, giving $\Psi(\psi) - \psi \le 0$. At its unique minimum $\psi_* \in (W_1, W_2)$ (where $\Psi'(\psi_*) = 1$), integrating $\Psi' - 1 \ge \rho_1 - 1$ from 0 and $\Psi' - 1 \le \beta$ to $\frac{1}{2}$ gives the two upper bounds. $\square$

Proposition 3.2 (Global C^8 symplectomorphism KS_τ)**.** *For* $\tau \in [0,1]$, *define* $KS_\tau \colon \mathbb{C} \to \mathbb{C}$ *via the generating function* $S(X, \psi; \tau) = X\psi + \tau M(X)(\Psi(\psi) - \psi)$ *about* $c_* = (-d, 0)$*:*

$$\phi = \psi + \tau M'(X)\big(\Psi(\psi) - \psi\big), \qquad X_n = X + \tau M(X)\big(\Psi'(\psi) - 1\big), \tag{3}$$

sending $w = c_* + \sqrt{X/\pi}\, e^{2\pi i\phi}$ *to* $w' = c_* + \sqrt{X_n/\pi}\, e^{2\pi i\psi}$, *with generator* $g_\tau(w) = \partial_\tau S = M(X)(\Psi(\psi) - \psi)$. *Then* $KS_\tau \in C^8(\mathbb{C}, \mathbb{C})$ *is a global symplectomorphism of* $(\mathbb{C}, dw_x \wedge dw_y)$ *with explicit inverse,* $g_\tau \in C^8(\mathbb{C}, \mathbb{R})$ *(being* C^9 *in* (X, ψ)*, while* $\psi(w)$ *is* C^8 *across* $X = X_w$ *where* $M'(X) = S_4(\min(X/X_w, 1)) \in C^8$*), and for any* C^k *(*$2 \le k \le 8$*) profile* $\tau = \tau(G) \in [0,1]$*, the suspension* $(G, v, w) \mapsto \big(G, v - \tau'(G) g_{\tau(G)}(w), KS_{\tau(G)}(w)\big)$ *is a* C^{k-1} *symplectomorphism of* $(\mathbb{R}^2 \times \mathbb{C}, dG \wedge dv + dw_x \wedge dw_y)$.

Proof. Since $\partial_\psi\phi = 1+\tau M'(X)(\Psi'(\psi)-1) \geq (1-\tau M') + \tau M'\rho_1 \geq \rho_1 > 0$ and $\partial_X X_n = 1+\tau(\Psi'(\psi)-1)M'(X) \geq \rho_1 > 0$ with $X_n(0)=0$ and $\lim_{X\to\infty} X_n = \infty$, both scalar equations have unique global roots on $\mathbb{R}/\mathbb{Z}$ and $[0,\infty)$ (for $X \geq X_w$, $M(X) = X - \frac{1}{2}X_w$ gives the closed-form inverse $X = (X_n + \frac{1}{2}\tau(\Psi'(\psi)-1)X_w)/(1+\tau(\Psi'(\psi)-1))$). As $X = \pi|w-c_*|^2 \to 0$, $S_4'(u) = O(u^8)$ implies $M'(X) = O(|w-c_*|^{18})$ and $M(X) = O(|w-c_*|^{20})$, so $KS_\tau(w) - w = O(|w-c_*|^{17})$ and $g_\tau(w) = O(|w-c_*|^{20})$. These polar expressions extend with at least C^8 regularity across $w = c_*$; together with the C^8 regularity across $X = X_w$ (where $\psi(w)$ inherits the C^8 regularity of $M'(X)$, and $\min(k-1,8) = k-1$ for $2 \leq k \leq 8$), this gives the claimed global C^8 regularity. Finally, $dS = \phi\,dX + X_n\,d\psi + g_\tau\,d\tau$ gives $dX_n \wedge d\psi - dX \wedge d\phi = -dg_\tau \wedge d\tau$, so substituting $\tau = \tau(G)$ yields $dG \wedge d(v - \tau' g_\tau) + dX_n \wedge d\psi = dG \wedge dv + dX \wedge d\phi$. □

3.2. **Full-disc quantitative nesting, hole clearance, and slit clearance.**

Theorem 3.3 (Quantitative keyhole bounds and the constant $8.4 > 2+2\pi$)**.** *Assume* (2). *Put* $x_0 = \pi r_h^2$, $\theta_0 = W_1$, *and*

$$\delta_{\mathrm{rel}} = \beta\pi(r_a+d)(r_a+8.4\,d), \qquad \delta_{\mathrm{small}} = 4\pi^3 d(r_a+d)W_2^2.$$

(i) (Disc nesting for all $\tau \in [0,1]$.) For every $P_0 \in [0,\pi r_a^2]$, every $\tau \in [0,1]$, and every $w \in \bar{D}(P_0) = \{z \in \mathbb{C} : \pi|z|^2 \leq P_0\}$, $w' = KS_\tau(w)$ satisfies

$$\pi|w'|^2 \leq \max\Big(P_0 + \max\big(\tau\,\delta_{\mathrm{rel}},\ \delta_{\mathrm{small}}\big),\ \pi d^2\Big). \tag{4}$$

In particular, whenever $P_0 \geq \pi d^2$, $\pi|w'|^2 \leq P_0 + \max(\tau\,\delta_{\mathrm{rel}}, \delta_{\mathrm{small}})$.

(ii) (Hole and slit clearance at $\tau = 1$.) For every $w \in \bar{D}(\pi r_a^2)$, $w' = KS_1(w)$ satisfies

$$\pi|w'|^2 > x_0 = \pi r_h^2 \qquad \text{and} \qquad |\theta(w')| > \theta_0 = W_1,$$

where $\theta(w') = \arg(w')/(2\pi) \in (-\frac{1}{2}, \frac{1}{2}]$; thus $KS_1(\bar{D}(\pi r_a^2))$ is disjoint from $\bar{D}(x_0)$ and from the closed sector $\{|\theta| \leq \theta_0\} \cup [0,\infty)$.

Proof. Let $w \in \bar{D}(P_0) \subset \bar{D}(\pi r_a^2)$. Then $R = |w-c_*| \leq |w|+d \leq r_a+d$ and $X = \pi R^2 \leq \pi(r_a+d)^2$. Write $w' = c_* + R_n e^{2\pi i\psi}$ ($X_n = \pi R_n^2$). By conjugation symmetry $\bar{w} \mapsto \overline{w'}$, assume $\psi \in [0,\frac{1}{2}]$; then Lemma 3.1 gives $0 \leq \phi \leq \psi \leq \frac{1}{2}$. From $|c_* + re^{2\pi i\alpha}|^2 = d^2 + r^2 - 2dr\cos(2\pi\alpha)$ we have

$$\pi|w'|^2 - \pi|w|^2 = (X_n - X) + 2\pi d(R-R_n)\cos(2\pi\psi) + 2\pi dR\big(\cos(2\pi\phi) - \cos(2\pi\psi)\big). \tag{5}$$

Proof of (i). Split $[0,\frac{1}{2}]$ by the sign of $\Psi'(\psi) - 1$.

Case A: $\Psi'(\psi) \leq 1$ (so $\psi \leq W_2 \leq \frac{1}{6}$, $\cos(2\pi\psi) \geq \frac{1}{2} > 0$, and $R_n \leq R$). If $R + R_n < 2d\cos(2\pi\psi)$, then $2R_n \leq R + R_n < 2d\cos(2\pi\psi)$, so $R_n < d\cos(2\pi\psi)$ and directly

$$|w'|^2 = d^2 + R_n\big(R_n - 2d\cos(2\pi\psi)\big) \leq d^2 \implies \pi|w'|^2 \leq \pi d^2.$$

If instead $R + R_n \geq 2d\cos(2\pi\psi)$, then $(X_n - X) + 2\pi d(R-R_n)\cos(2\pi\psi) = \pi(R - R_n)\big(2d\cos(2\pi\psi) - (R+R_n)\big) \leq 0$. Moreover, since $0 \leq \phi \leq \psi \leq W_2$, $\cos(2\pi\phi) - \cos(2\pi\psi) \leq 1 - \cos(2\pi\psi) = 2\sin^2(\pi\psi) \leq 2\pi^2\psi^2 \leq 2\pi^2 W_2^2$. Hence (5) gives $\pi|w'|^2 - \pi|w|^2 \leq 2\pi d(r_a + d)(2\pi^2 W_2^2) = 4\pi^3 d(r_a+d)W_2^2 = \delta_{\mathrm{small}}$.

Case B: $\Psi'(\psi) > 1$ (so $1 < \Psi'(\psi) \leq 1+\beta$ and $R_n \geq R$). First, $X_n - X = \tau M(X)(\Psi'(\psi)-1) \leq \tau\beta X \leq \tau\beta\pi(r_a+d)^2$. Second, if $\cos(2\pi\psi) \geq 0$ then $2\pi d(R-R_n)\cos(2\pi\psi) \leq 0$; if $\cos(2\pi\psi) < 0$ then $R_n - R = (X_n - X)/(\pi(R_n+R)) \leq \tau\beta\pi R^2/(2\pi R) = \frac{1}{2}\tau\beta R$, so $2\pi d(R-R_n)\cos(2\pi\psi) \leq 2\pi d(R_n - R) \leq \tau\beta\pi d(r_a+d)$. Third, by Lemma 3.1, $0 \leq \psi - \phi = \tau M'(X)(\psi - \Psi(\psi)) \leq \frac{1}{2}\tau\beta$, so $\cos(2\pi\phi) - \cos(2\pi\psi) \leq 2\pi(\psi-\phi) \leq \pi\tau\beta$, giving $2\pi dR(\cos(2\pi\phi) - \cos(2\pi\psi)) \leq 2\pi^2\tau\beta d(r_a+d) = (2\pi)\tau\beta\pi d(r_a+d)$. Summing the three terms in (5) yields

$$\pi|w'|^2 - \pi|w|^2 \leq \tau\beta\pi(r_a+d)\big[(r_a+d) + d + 2\pi d\big] = \tau\beta\pi(r_a+d)\big[r_a + (2+2\pi)d\big] \leq \tau\,\delta_{\mathrm{rel}},$$

since $2+2\pi = 8.283185\cdots < 8.4$. Combining Cases A and B proves (4).

Proof of (ii). Set $\tau = 1$. If $\psi \in [0, W_1]$, then $\Psi'(\psi) = \rho_1$ and $M(X) \geq X - \frac{1}{2}X_w$, so by (2c),

$$X_n = X - (1-\rho_1)M(X) \leq \rho_1 X + \tfrac{1}{2}(1-\rho_1)X_w < \rho_1\pi(d+r_a)^2 + \tfrac{1}{2}X_w < \pi(d-r_h)^2.$$

Hence $R_n < d - r_h$, which places w' strictly in the left half-plane $\mathrm{Re}(w') = -d + R_n\cos(2\pi\psi) < -r_h < 0$; consequently $|w'| \geq |\mathrm{Re}(w')| > r_h$ ($\pi|w'|^2 > x_0$) and $|\theta(w')| > \frac{1}{4} > \frac{1}{6} \geq W_2 > W_1$. If instead $\psi \in (W_1, \frac{1}{2}]$, then for $\psi \geq \frac{1}{4}$ we again have $\mathrm{Re}(w') \leq -d < -r_h$, whereas for $\psi \in (W_1, \frac{1}{4})$ the ray $\{c_* + te^{2\pi i\psi} : t \geq 0\}$ has perpendicular distance $d\sin(2\pi\psi) > d\sin(2\pi W_1) \geq r_h$ from the origin by (2b), so $|w'| > r_h$ for all $R_n \geq 0$; moreover, whenever $\mathrm{Re}(w') > 0$ (so $R_n\cos(2\pi\psi) > d$), $\tan(2\pi\theta(w')) = \frac{R_n\sin(2\pi\psi)}{R_n\cos(2\pi\psi)-d} > \tan(2\pi\psi) > \tan(2\pi W_1)$, whence $\theta(w') > \psi > W_1 = \theta_0$. □

4. Cell layout and the four-dimensional keyhole comparator

4.1. **The flat model, fundamental domain, and cell embeddings C_i.** Let $k \geq 10$ and $\eta \in (0, 1-\sqrt{99/100}]$, and set $s = \sqrt{k} > 3$, $t_0 = 1 - \frac{1}{2}\eta$, $a = t_0/s$, $h = t_0/3$, $c_s = 3s - 9 > 0$, $g = 3(1-t_0) = \frac{3}{2}\eta$, and $T = g/c_s$. On the flat chart $(G, v, w) \in \mathbb{R}\times\mathbb{R}\times\mathbb{C}$ with symplectic form $\Omega_{\mathrm{fl}} = dG\wedge dv + dw_x \wedge dw_y$, define $P(p) = 3 - 9p$, $J(p) = 3sp - 3$, and $K(p) = c_s p$ for $p = \pi|w|^2 < \frac{1}{3}$. Notice the exact period identity

$$K(p) - J(p) = (3s-9)p - (3sp-3) = 3 - 9p = P(p) > 0, \tag{6}$$

and the deck group generated by the two C^ω symplectomorphisms

$$T_u(G, v, w) = \big(G + P(p),\, v,\, we^{2\pi i(9v)}\big), \qquad T_v(G, v, w) = (G, v+1, w),$$

which commute modulo integer fibre turns ($T_uT_v = T_vT_u$ because $e^{2\pi i\cdot 9} = 1$) and has fundamental domain $\mathcal{F}_{\mathrm{fl}} = \{p < \frac{1}{3},\ J(p) \leq G < K(p),\ 0 \leq v < 1\}$ (for T_u alone, the fundamental slice is $\mathcal{F}_u = \{p < \frac{1}{3},\ J(p) \leq G < K(p)\}\times\mathbb{R}$). Define the level curves

$$O(G) = \frac{G+3}{3s}, \quad E_{\mathrm{lin}}(G) = \frac{G}{c_s}, \quad P_{\mathrm{top}}(G) = \frac{ka - 3 - G}{k - 3s}, \quad \mathrm{slack}(G) = T\Big(1 + \frac{\min(G,0)}{g}\Big).$$

Direct algebra gives the linear comparison identity of Paper I in level-G form:

$$O(G) - E_{\mathrm{lin}}(G) - P_{\mathrm{top}}(G) \equiv T \qquad (\forall\, G \in \mathbb{R}), \tag{7}$$

and $p < O(G) \iff G > J(p)$, while $p > E_{\mathrm{lin}}(G) \iff G < K(p)$.

Lemma 4.1 (Cell-to-flat embeddings C_i)**.** *For $0 \leq i \leq k-1$, define $C_i\colon \mathcal{C}_a \to \mathbb{R}\times\mathbb{R}\times\mathbb{C}$ on $(X, T_{\mathrm{cell}}, w)$ ($p = \pi|w|^2 < a$, $0 < X < a - p$, $0 < T_{\mathrm{cell}} < 1$) by*

$$C_i(X, T_{\mathrm{cell}}, w) = \begin{cases} \big(J(p) + 9X,\ \frac{i-4+T_{\mathrm{cell}}}{9},\ we^{-2\pi i(3sT_{\mathrm{cell}}/9)}\big), & 0 \leq i < 9, \\ \big(J(p) + i(a-p) + X,\ T_{\mathrm{cell}} + \frac{1}{2},\ we^{-2\pi i((3s-i)T_{\mathrm{cell}})}\big), & 9 \leq i \leq k-1. \end{cases}$$

Then each C_i is a real-analytic symplectic embedding ($C_i^*\Omega_{\mathrm{fl}} = dX\wedge dT_{\mathrm{cell}} + \omega_V$). The nine group images $\mathcal{G}_i = C_i(\mathcal{C}_a)$ ($0 \leq i < 9$) are pairwise disjoint in the T_u-slice $\mathcal{F}_u$ (lying in 1-periodically disjoint v-windows $(\frac{i-4}{9}, \frac{i-3}{9}) \subset (-\frac{4}{9}, \frac{5}{9})$ of total width 1) and satisfy

$$J(p) < G < J(p) + 9(a-p) = (3s-9)p + 9a - 3 < 3sa - 3 = -g \qquad (\forall\,(G, v, w) \in \mathcal{G}_i). \tag{8}$$

The stack images $\mathcal{S}_i = C_i(\mathcal{C}_a)$ ($9 \leq i < k$) are pairwise disjoint (separated by the hyperplanes $A_i(p) = J(p) + i(a-p)$), satisfy $G > J(p) + 9(a-p)$, $G < J(0) + ka = st_0 - 3 =: G_{\max}$, and at each level G have fibre slice $p \leq P_{\mathrm{top}}(G)$ (with $p < a$).

Proof. For $i<9$, $d(J(p)+9X)\wedge d(\frac{i-4+T_{\mathrm{cell}}}{9})=(3s\,dp+9\,dX)\wedge\frac{dT_{\mathrm{cell}}}{9}=dX\wedge dT_{\mathrm{cell}}+\frac{s}{3}dp\wedge dT_{\mathrm{cell}}$, which cancels $dp\wedge d(\theta-\frac{s}{3}T_{\mathrm{cell}})-dp\wedge d\theta$; similarly for $i\ge 9$, $d(J(p)+i(a-p)+X)\wedge dT_{\mathrm{cell}}=dX\wedge dT_{\mathrm{cell}}+(3s-i)dp\wedge dT_{\mathrm{cell}}$. Since $3s-9>0$ and $p<a$, $(3s-9)p+9a-3<(3s-9)a+9a-3=3sa-3=3t_0-3=-g$. Finally, $G<J(p)+k(a-p)=-3+ka-(k-3s)p$ is equivalent to $p<P_{\mathrm{top}}(G)$. □

4.2. **The four-dimensional keyhole comparator $F=F_2\circ F_1$.** Fix $0<\lambda_2<\lambda_a<1$ and $\mu\in(0,1)$ ($\lambda_a=\frac{9}{10}$, $\lambda_2=\frac{3}{10}$, $\mu=\frac{1}{2}$), and put $w_G=\lambda_a g$ and $w_2=\lambda_2 g$. Let K_C be a smooth keyhole on $\bar D(a)$ satisfying (2) with hole $x_0=\pi r_h^2$ and slit half-width $\theta_0=W_1$. Define

$$\tau(G)=S_4\Big(\frac{G+w_G}{w_G-w_2}\Big),\quad c(G)=\frac{w_2}{c_s}R_4\Big(\frac{G+w_2}{w_2}\Big),\quad \vartheta(\theta)=\theta-\tfrac{1}{2}\operatorname{sgn}(\theta)\ \ \big(\theta\in(-\tfrac12,\tfrac12]\setminus\{0\}\big).$$

Notice that $\vartheta(\arg(w)/(2\pi))=\arg(-w)/(2\pi)\in(-\frac12,\frac12)$ is real-analytic on the slit plane $\mathbb{C}\setminus[0,\infty)$ (in particular across the negative real axis $\theta=\pm\frac12$). Define $F_1(G,v,w)=\big(G,v-\tau'(G)g_{\tau(G)}(w),KS_{\tau(G)}(w)\big)$ and, on $\{c(G)=0\}\cup\{(G,v,w_1):w_1\in\mathbb{C}\setminus[0,\infty),\ \pi|w_1|^2+c(G)>0\}$, define $F_2(G,v_1,w_1)=(G,v_1,w_1)$ when $c(G)=0$ ($G\le -w_2$) and

$$F_2(G,v_1,w_1)=\Big(G,\ v_1-c'(G)\vartheta(\theta_1),\ \sqrt{\frac{x_1+c(G)}{\pi}}\,e^{2\pi i\theta_1}\Big)\qquad (x_1=\pi|w_1|^2,\ \theta_1=\arg(w_1)/(2\pi))$$

when $c(G)>0$ ($G>-w_2$). Set $F=F_2\circ F_1$, $F^{(i)}=\mathrm{id}$ for $0\le i<9$, and $F^{(i)}=F$ for $9\le i<k$.

Theorem 4.2 (Domain, symplecticity, and stack-corridor separation of F)**.** *Suppose K_C satisfies (2) on $\bar D(a)$ together with the schedule inequalities:*

- *(H6)* $\max\big(S_4(u)\delta_{\mathrm{rel}},\delta_{\mathrm{small}}\big)\le(1-\mu)T\big(1-\lambda_a+u(\lambda_a-\lambda_2)\big)$ *for $u\in[0,1]$;*
- *(H7)* $\max(\delta_{\mathrm{rel}},\delta_{\mathrm{small}})+\lambda_2TR_4(u)\le(1-\mu)T\big(1-\lambda_2(1-u)\big)$ *for $u\in[0,1]$;*
- *(H8)* $\pi d^2\le T(1-\mu-\frac12\lambda_2)$ *and* $\pi d^2+\frac12\lambda_2T+\mu T\le O(-w_G)$.

Then on the open stack union $\mathcal{S}=\bigcup_{i=9}^{k-1}\mathcal{S}_i$:

(i) For every $(G,v,w)\in\mathcal{S}$ with $G>-w_2$, $\tau(G)=1$ and $w\in B^\circ(a)$, so by Theorem 3.3(ii) $w_1=KS_1(w)$ satisfies $x_1>x_0>0$ and $|\theta_1|>\theta_0>0$, avoiding $[0,\infty)$ by an open sector. As $G\downarrow -w_2$, $c(G)$ and $c'(G)$ vanish to order 9 while $|\vartheta(\theta_1)|<\frac12$ and $x_1>x_0$. Hence F is a C^7 symplectic embedding on an open neighbourhood of $\mathcal{S}$, with explicit C^7 inverse $F_1^{-1}\circ F_2^{-1}$.

(ii) Every image point $(G,v_2,w_2)\in F(\mathcal{S})$ ($p_2=\pi|w_2|^2$) lies strictly inside a single fundamental domain $[J(p_2),K(p_2))$ of T_u and satisfies:

$$p_2<h_K\qquad (\text{whenever } h_K\ge a \text{ and } h_K>\max_{G\in[-w_G,G_{\max}]}(O(G)-\mu\,\mathrm{slack}(G))), \tag{9}$$

$$J(p_2)<G<K(p_2)-\min\big(w_2,\ c_sx_0+\tfrac12w_2\big)\le K(p_2)-\tfrac12w_2, \tag{10}$$

$$G>J(p_2)+9(a-p_2)\qquad \text{whenever } p_2<a. \tag{11}$$

Consequently, the k sets $\mathcal{G}_0,\dots,\mathcal{G}_8,F(\mathcal{S}_9),\dots,F(\mathcal{S}_{k-1})$ are pairwise disjoint modulo the deck group $\langle T_u,T_v\rangle$, and all of them miss the corridor $\mathcal{K}_{\mathrm{free}}=\{p<\frac13:\ K(p)-\frac12w_2\le G<K(p)\}\pmod{P(p)}$.

Proof. (i) F_1 is a global C^7 symplectomorphism by Proposition 3.2 ($\tau'\in C^7$). On $\{G>-w_2\}$, $d(x_1+c(G))\wedge d\theta_1+dG\wedge d(v_1-c'(G)\vartheta(\theta_1))=dx_1\wedge d\theta_1+c'(G)dG\wedge d\theta_1+dG\wedge dv_1-c'(G)dG\wedge d\vartheta(\theta_1)=dG\wedge dv_1+dx_1\wedge d\theta_1$ since $\vartheta'(\theta_1)=1$ off $\theta_1=0$. Because $\mathcal{S}_9,\dots,\mathcal{S}_{k-1}$ are pairwise disjoint and F is injective on their union, $F(\mathcal{S}_9),\dots,F(\mathcal{S}_{k-1})$ are pairwise disjoint.

(ii) If $G\le -w_G$, $F=\mathrm{id}$, so $p_2=p<a\le h_K$, $J(p_2)+9(a-p_2)<G\le -w_G<-w_2<K(p_2)-w_2$. Now suppose $G\in(-w_G,G_{\max})$. Since $w\in B^\circ(\min(a,P_{\mathrm{top}}(G)))$, Theorem 3.3(i)

gives $x_1 \le \max(P_{\mathrm{top}}(G) + \max(\tau(G)\delta_{\mathrm{rel}}, \delta_{\mathrm{small}}), \pi d^2)$, and adding $c(G)$ gives $p_2 = x_1 + c(G)$. On $(-w_G, -w_2]$ ($c(G) = 0$), (H6) gives $P_{\mathrm{top}}(G) + \max(\tau\delta_{\mathrm{rel}}, \delta_{\mathrm{small}}) \le P_{\mathrm{top}}(G) + (1-\mu)\,\mathrm{slack}(G)$. Notice that for $G \le 0$, $G/c_s = T(G/g)$, so (7) gives the exact identity

$$O(G) - P_{\mathrm{top}}(G) = T + \frac{G}{c_s} = T\Big(1 + \frac{G}{g}\Big) = \mathrm{slack}(G) \qquad (G \le 0),$$

whereas for $G \ge 0$, $O(G) - P_{\mathrm{top}}(G) = T + E_{\mathrm{lin}}(G)$ with $\mathrm{slack}(G) = T$. Therefore:

- On $(-w_G, -w_2]$, (H6) gives $P_{\mathrm{top}}(G) + \max(\tau\delta_{\mathrm{rel}}, \delta_{\mathrm{small}}) \le O(G) - \mu\,\mathrm{slack}(G)$;
- On $[-w_2, 0]$, (H7) gives $P_{\mathrm{top}}(G) + \max(\delta_{\mathrm{rel}}, \delta_{\mathrm{small}}) + c(G) \le O(G) - \mu\,\mathrm{slack}(G)$;
- On $[0, G_{\max}]$, $c(G) = \frac{G}{c_s} + \frac{1}{2}\lambda_2 T$, and (H7) at $u = 1$ gives $\max(\delta_{\mathrm{rel}}, \delta_{\mathrm{small}}) + \frac{1}{2}\lambda_2 T \le (1-\mu)T$, so $P_{\mathrm{top}}(G) + \max(\delta_{\mathrm{rel}}, \delta_{\mathrm{small}}) + c(G) \le P_{\mathrm{top}}(G) + \frac{G}{c_s} + (1-\mu)T = O(G) - \mu T = O(G) - \mu\,\mathrm{slack}(G)$.

In the remaining $\pi d^2 + c(G)$ branch of (4): on $(-w_G, 0]$, $c(G) \le c(0) = \frac{1}{2}\lambda_2 T$ and $\mathrm{slack}(G) \le T$, so (H8) gives $\pi d^2 + c(G) + \mu\,\mathrm{slack}(G) \le \pi d^2 + \frac{1}{2}\lambda_2 T + \mu T \le O(-w_G) < O(G)$; on $[0, G_{\max}]$, $O(G) - \frac{G}{c_s} = P_{\mathrm{top}}(G) + T \ge T$, so (H8) gives $\pi d^2 + c(G) + \mu T = \pi d^2 + \frac{G}{c_s} + (\frac{1}{2}\lambda_2 + \mu)T \le \frac{G}{c_s} + T \le O(G)$. Thus $p_2 \le O(G) - \mu\,\mathrm{slack}(G) < O(G)$ holds for *all* $G \in (-w_G, G_{\max})$, which gives both (9) and $G > 3sp_2 - 3 = J(p_2)$.

For the upper bound (10): if $G \in (-w_G, -w_2]$, then $K(p_2) \ge 0$, so $K(p_2) - G \ge -G \ge w_2$. If $G \in (-w_2, G_{\max})$, Theorem 3.3(ii) gives $x_1 > x_0$, so $p_2 > x_0 + c(G)$; writing $u = (G + w_2)/w_2 > 0$ ($G = (u-1)w_2$) and using $u - R_4(u) \le \lim_{t\to\infty}(t - R_4(t)) = \frac{1}{2}$, we have

$$K(p_2) - G > c_s x_0 + c_s c(G) - G = c_s x_0 + w_2\big(1 - (u - R_4(u))\big) \ge c_s x_0 + \tfrac{1}{2} w_2.$$

Finally, for $G > -w_G = -0.9g$ and any $p_2 < a$, (8) gives $J(p_2) + 9(a - p_2) < -g < -0.9g < G$, proving (11) and disjointness modulo $\langle T_u, T_v\rangle$. □

5. Marked-fibre twists and corridor routing

In the flat chart, put $u = G/P(p)$ ($P(p) = 3 - 9p > 0$). The three straight marked fibres of the cubic transfer (Section 6) are $\mathcal{M}_1 = \{(u,v) \equiv (\frac{5}{6}, 0)\}$, $\mathcal{M}_2 = \{(u,v) \equiv (\frac{1}{3}, 0)\}$, and $\mathcal{M}_3 = \{(u,v) \equiv (0, \frac{1}{2})\}$ modulo $\mathbb{Z}^2$. Since the ball images fill (u,v) across (u_j, v_j), we interpose two closed-form C^4 deck-equivariant symplectomorphisms $\Psi = \Phi_g \circ \Psi_S$ that pull all three marked fibres back into the empty corridor $\mathcal{K}_{\mathrm{free}}$.

Construction 5.1 (Toric twists Ψ_S and Φ_g)**.** Fix constants $a_g = \frac{1}{10}$, $(\omega_{m,0}, \omega_{m,1}) = (0.015, 0.045)$, $(\omega_{u,0}, \omega_{u,1}) = (0.02, 0.12)$, $v_{\mathrm{comp}} = \frac{1}{4}$, $(\omega_{c,0}, \omega_{c,1}) = (0.05, 0.14)$, $\delta = \frac{1}{4}w_2$, $u_* = -c_s/9$, $c_0 = \delta - c_s/3$, $n_j = -\,\mathrm{round}(u_j - u_*) \in \mathbb{Z}$, $\phi_j = u_j + n_j - u_*$, and shifted v-targets $(v_1', v_2', v_3') = (-a_g, 0, \frac{1}{2})$. Using $m = 2$ plateau bumps $\beta_v = \beta_{\omega_{m,0},\omega_{m,1},2}$, $\beta_c = \beta_{\omega_{c,0},\omega_{c,1},2}$ (mass $M_c = \omega_{c,0} + \omega_{c,1}$), and $\beta_u = \beta_{\omega_{u,0},\omega_{u,1},2}$:

(1) For $v \in \mathbb{R}$, reduce $\tilde v \in [-\frac{1}{3}, \frac{2}{3})$ modulo 1 and set $b_j(v) = \beta_v(\tilde v - v_j') - \frac{M_v}{M_c}\beta_c(\tilde v - v_{\mathrm{comp}})$, $B_j(v) = \mathcal{B}_v(\tilde v - v_j') - \frac{M_v}{M_c}\mathcal{B}_c(\tilde v - v_{\mathrm{comp}})$ (so $B_j(v)$ is 1-periodic in $C^5(\mathbb{R})$ because $\int_{-1/3}^{2/3} b_j = 0$). Put $S_v(p,v) = c_0 + P(p)\sum_{j=1}^3 \phi_j b_j(v)$, $S_p(v) = -9\sum_{j=1}^3 \phi_j B_j(v)$, and

$$\Psi_S(G, v, w) = \big(G + S_v(p,v),\ v,\ we^{-2\pi i S_p(v)}\big).$$

(2) For $u \in \mathbb{R}$, reduce $\tilde u \in [\frac{1}{5}, \frac{6}{5})$ modulo 1 and set $g_{\mathrm{tw}}(u) = -a_g\big(\beta_u(\tilde u - \frac{5}{6}) - \beta_u(\tilde u - \frac{7}{12})\big)$ and $G_g(u) = -a_g\big(\mathcal{B}_u(\tilde u - \frac{5}{6}) - \mathcal{B}_u(\tilde u - \frac{7}{12})\big)$ (1-periodic in $C^4(\mathbb{R})$ and $C^5(\mathbb{R})$). Define

$$\Phi_g(G, v, w) = \big(G,\ v - g_{\mathrm{tw}}(u),\ we^{2\pi i \cdot 9(G_g(u) - u g_{\mathrm{tw}}(u))}\big) \qquad (u = G/P(p)).$$

Proposition 5.2 (Symplecticity, deck equivariance, and marked-fibre routing)**.** *On* $\{p < \frac{1}{3}\}$*:*

(i) Ψ_S *and* Φ_g *are* C^4 *symplectomorphisms preserving* $p=\pi|w|^2$*, with closed-form inverses obtained by reversing the sign of* (S_v,S_p) *and* g_{tw}*, and both commute with* T_u *and* T_v*.*

(ii) Under the window-separation checks (H10) (the four v*-intervals* $[v_j'\pm 0.045]$*,* $[0.25\pm 0.14]$ *are pairwise disjoint in* $[-\frac{1}{3},\frac{2}{3})$*, and the two* u*-lobes* $[\frac{5}{6}\pm 0.12]$*,* $[\frac{7}{12}\pm 0.12]$ *are disjoint in* $[\frac{1}{5},\frac{6}{5})$ *and miss* $\frac{1}{3}$ *and* 1*), the preimage of the deck orbit of* $\mathcal{M}_j$ *under* $\Psi=\Phi_g\circ\Psi_S$ *is the exact spiral*

$$\Psi^{-1}\big(\mathrm{deck}(\mathcal{M}_j)\big)=\big\{(G,v,w): G\equiv K(p)-\delta \pmod{P(p)},\ v\equiv v_j' \pmod 1\big\}. \tag{12}$$

Since $0<\delta=\frac{1}{4}w_2<\frac{1}{2}w_2$ *and* $3-9a>\delta$ *(H9b), all three spirals lie inside the free corridor* $\mathcal{K}_{\mathrm{free}}$ *and are disjoint from* $\mathcal{G}_0,\dots,\mathcal{G}_8,F(\mathcal{S}_9),\dots,F(\mathcal{S}_{k-1})$*.*

Proof. (i) Writing $S(p,v)=c_0v+P(p)\sum_j\phi_jB_j(v)$ locally, $\partial_vS=S_v$ and $\partial_pS=S_p$, so $d(G+S_v)\wedge dv+dp\wedge d(\theta-S_p)=dG\wedge dv+dp\wedge d\theta$. For Φ_g, writing $H_g(G,p)=P(p)G_g(G/P(p))$ gives $\partial_GH_g=g_{\mathrm{tw}}(u)$ and $\partial_pH_g=-9(G_g(u)-ug_{\mathrm{tw}}(u))$, so Φ_g is the exact time-one flow of H_g (constant along orbits since G and p are preserved) and is C^4 because $g_{\mathrm{tw}}\in C^4$. Deck equivariance holds because b_j,B_j are 1-periodic in v and g_{tw},G_g are 1-periodic in u (for T_u on Φ_g: shifting $u\mapsto u+1$ shifts $ug_{\mathrm{tw}}(u)$ by $g_{\mathrm{tw}}(u)$, shifting the fibre phase by $-9g_{\mathrm{tw}}(u)$, which matches T_u's phase $9(v-g_{\mathrm{tw}}(u))$).

(ii) At $u_1\equiv\frac{5}{6}$, $g_{\mathrm{tw}}(\frac{5}{6})=-a_g(1-0)=-a_g$, whereas at $u_2\equiv\frac{1}{3}$ and $u_3\equiv 0$, $g_{\mathrm{tw}}=0$ by (H10). Thus $\Phi_g^{-1}(\mathrm{deck}(\mathcal{M}_j))=\{u\equiv u_j \pmod 1,\ v\equiv v_j' \pmod 1\}$. At $v\equiv v_j'\pmod 1$, (H10) gives $b_j(v_j')=1$, $b_l(v_j')=0$ $(l\neq j)$, and $\beta_c(v_j'-v_{\mathrm{comp}})=0$, so $S_v(p,v_j')=c_0+P(p)\phi_j$. Since $\phi_j=u_j+n_j+c_s/9$ and $c_0=\delta-c_s/3$,

$$S_v(p,v_j')=(3-9p)\Big(u_j+n_j+\frac{c_s}{9}\Big)+\delta-\frac{c_s}{3}=P(p)(u_j+n_j)-c_sp+\delta=P(p)(u_j+n_j)-\big(K(p)-\delta\big).$$

Hence $G+S_v(p,v_j')\equiv P(p)u_j \pmod{P(p)}$ if and only if $G\equiv K(p)-\delta \pmod{P(p)}$, proving (12). Since $0<\delta=\frac{1}{4}w_2<\min(\frac{1}{2}w_2,3-9a)$, in the fundamental domain $[J(p),K(p))$ we have $J(p)+9(a-p)=K(p)-(3-9a)<K(p)-\delta$ (for $p<a$) and $\sup F(\mathcal{S})_G<K(p)-\frac{1}{2}w_2<K(p)-\delta$ by (10). □

6. Closed-form transfer to the four-ball

6.1. **Post chart, corner bands, and simplex gluing.** On $\{p<h_K\}$ $(0<h_K<\frac{1}{3}$, $D=1-3h_K>0)$, the post chart $\Pi_{\mathrm{post}}(G,v,w)=(\sigma,\psi,w_{\mathrm{post}})$ with $\psi_0=-\frac{1}{6}$ and $\ell=1$ sets $u_a=G/P(p)+1$, $\sigma=P(p)v$, $\psi=-\frac{1}{6}-u_a$, and $\vartheta=\theta-9(u_a-1)v$ $(w_{\mathrm{post}}=\sqrt{p/\pi}\,e^{2\pi i\vartheta})$. Direct differentiation gives $d\sigma\wedge d\psi+dp\wedge d\vartheta=(P\,dv-9v\,dp)\wedge(-P^{-1}dG-9GP^{-2}dp)+dp\wedge(d\theta-9\frac{G}{P}dv-9v\frac{dG}{P}-81\frac{Gv}{P^2}dp)=dG\wedge dv+dp\wedge d\theta$. Under Π_{post}, T_u becomes $\tau_u\colon\psi\mapsto\psi-1$, T_v becomes $\tau_v\colon(\sigma,\psi,p,\vartheta)\mapsto(\sigma+P(p),\psi,p,\vartheta+9\psi+\frac{1}{2})$ (because $-9\psi_0=\frac{3}{2}\equiv\frac{1}{2}\pmod 1$), and the three straight marked fibres $(\mathcal{M}_1,\mathcal{M}_2,\mathcal{M}_3)$ become

$$(v_c,\psi)\equiv(0,0),\quad(0,\tfrac{1}{2}),\quad(\tfrac{1}{2},-\tfrac{1}{6})\qquad\big(v_c=\sigma/P(p),\ \psi \bmod 1\big).$$

Fix fractions $(f_m,f_{d_2},f_{\sigma_0},f_\delta)=(0.32,0.20,0.20,0.10)$ of D, so $m=0.32D$, $d_2=0.20D$, $\sigma_0=0.20D$, and $\delta_{\mathrm{kh}}=0.10D$, and let K_B be the corner keyhole on $\bar{D}(h_K)$ with budget $0.8\delta_{\mathrm{kh}}$. Set $(\kappa_0,\kappa_1,\kappa_2)=(-\frac{1}{12},\frac{1}{12},0)$, phase offsets $\mathrm{off}_0=(0,0)$, $\mathrm{off}_1=(\frac{1}{12},\frac{1}{4})$, $\mathrm{off}_{2s}=(\frac{1}{12},\frac{1}{4})$, $\mathrm{off}_{2c}=(\frac{1}{12},0)$, $\mathrm{left}_0=\mathrm{off}_{2c}$, $\mathrm{left}_1=\mathrm{off}_0$, $\mathrm{left}_2=\mathrm{off}_1$, and $\kappa=e^{-2\pi i/12}=e^{-i\pi/6}$.

Reduce $(\sigma,\psi,p,\vartheta)$ by τ_v^{-n} $(n=\lfloor v_c+\frac{1}{6}\rfloor)$ into the fundamental strip $v_c\in[-\frac{1}{6},\frac{5}{6})$, and put $c_j=\sigma-j(1-3p)$ for $j\in\{0,1,2\}$. Under the split inequality $2m<D-3\delta_{\mathrm{kh}}$ $(0.64<0.70)$, for any $p_2<h_K+\delta_{\mathrm{kh}}$ we have $1-3p_2>D-3\delta_{\mathrm{kh}}>2m$, so the three bands $\{|c_j|<m\}$ $(j=0,1,2)$ are pairwise disjoint both before and after the corner keyhole.

(1) *Corner keyhole $\mathcal{B}$ (**Band**).* Outside $\bigcup_{j=0}^{2}\{|c_j| < m\}$, $\mathcal{B} = \mathrm{id}$. On $\{|c_j| < m\}$, put $\tau_B(c) = 1 - S_4((|c| - d_2)/(m - d_2))$, $k_j(c) = \kappa_j S_4((c+m)/(2m))$, $\psi_1 = \psi + a_j + k_j(c)$, $v_j = \vartheta + b_j + 3jk_j(c) - 3j\psi_1$ ($(a_j, b_j) = \mathrm{left}_j$), apply $(w_2, g_B) = K_B^{\tau_B(c)}(\sqrt{p/\pi}e^{2\pi i v_j})$, set $\psi_2 = \psi_1 - \tau_B'(c)g_B$, $p_2 = \pi|w_2|^2$, $v_{j,2} = \arg(w_2)/(2\pi)$, recover $\sigma_{\mathrm{out}} = c + j(1 - 3p_2)$ (preserving c_j), $\psi_{\mathrm{out}} = \psi_2 - a_j - k_j(c)$, $\vartheta_{\mathrm{out}} = v_{j,2} + 3j\psi_2 - b_j - 3jk_j(c)$, and restore τ_v^n.

(2) *Simplex gluing $\mathcal{W}$.* On the output of $\mathcal{B}$ (reduced to $v_c \in [-\frac{1}{6}, \frac{5}{6})$):

- On the four sector pieces outside $\bigcup_{j=0}^{2}\{|c_j| < m\}$ ($0 \le v_c < \frac{1}{3}$, $\frac{1}{3} \le v_c < \frac{2}{3}$, $\frac{2}{3} \le v_c < \frac{5}{6}$, and $-\frac{1}{6} \le v_c < 0$, indexed by $s \in \{0, 1, 2s, 2c\}$), set $(\tilde\psi, \tilde\vartheta) = (\psi, \vartheta) + \mathrm{off}_s$ and define $(X_0, X_1, X_2; T_1, T_2)$ by

$$\begin{aligned}
s = 0: &\quad (1 - 2p - \sigma,\ p + \sigma,\ p;\ \tilde\psi,\ \tilde\vartheta - \tilde\psi),\\
s = 1: &\quad (p,\ 2 - 5p - \sigma,\ 4p + \sigma - 1;\ -\tilde\vartheta + 4\tilde\psi,\ -\tilde\vartheta + 5\tilde\psi),\\
s = 2s: &\quad (7p + \sigma - 2,\ p,\ 3 - 8p - \sigma;\ \tilde\vartheta - 8\tilde\psi,\ -\tilde\psi),\\
s = 2c: &\quad (1 - 2p + \sigma,\ p,\ p - \sigma;\ \tilde\vartheta + \tilde\psi,\ -\tilde\psi),
\end{aligned}$$

  and $W = (\sqrt{X_0}, \sqrt{X_1}e^{2\pi i T_1}, \sqrt{X_2}e^{2\pi i T_2}) \in S^5 \subset \mathbb{C}^3$.
- On band $j \in \{0, 1, 2\}$ ($|c| < m$, $c = c_j$), put $q(c) = \sigma_0(1 - S_4(|c|/d_2))$, $S(c) = \sqrt{c^2 + q(c)^2}$, $\lambda_+(c) = \frac{1}{2}(S(c) + c) \ge 0$, $\lambda_-(c) = \frac{1}{2}(S(c) - c) \ge 0$ (evaluated without cancellation via $q(c)^2/(2(S(c)+|c|))$), $\lambda(c) = \frac{1}{2}(c - S(c)) = -\lambda_-(c)$, $v_\ell = \mathrm{frac}(\vartheta + b_j + 3jk_j(c) - 3j(\psi + a_j + k_j(c))) \in (0, 1)$ (since $|c| \le d_2 \implies \tau_B(c) = 1 \implies v_\ell \notin \{0, 1\}$ on band interiors where $\lambda' \ne 0, 1$), $x_a = p + \lambda_+(c)$, $x_b = p + \lambda_-(c)$, $x_r = 1 - x_a - x_b$, $T_a = \psi + a_j + k_j(c) + \lambda'(c)v_\ell$, and permute $(\sqrt{x_r}, \sqrt{x_a}e^{2\pi i T_a}, \sqrt{x_b}e^{2\pi i(v_\ell - T_a)})$ by the cyclic shift $(j, j+1, j+2)$ mod 3.

(3) *Rotated ball chart χ_κ.* Set $Z_0 = (W_2 - \kappa W_1)/\sqrt{2}$, $Z_1 = (W_1 + \bar\kappa W_2)/\sqrt{2}$, $Z_2 = W_0$ (unitary rotation in (W_1, W_2), so $|Z|^2 = |W|^2 = 1$). For $Z_0 \ne 0$, set $\chi_\kappa([Z]) = \frac{\bar Z_0}{\sqrt{\pi}|Z_0|}(Z_1, Z_2) \in B^\circ(1) \subset \mathbb{C}^2 \cong \mathbb{R}^4$.

Theorem 6.1 (Global regularity, symplecticity, injectivity, and line avoidance of K_{post})**.** *Suppose $0 < h_K < \frac{1}{3}$, K_B satisfies (2) on $\bar D(h_K)$ with $\max(\pi d_B^2, \delta_{\mathrm{rel},B}, \delta_{\mathrm{small},B}) \le \delta_{\mathrm{kh}}$ (H13), $0 < d_2 < m$, $2m < D - 3\delta_{\mathrm{kh}}$, $\eta_1 := \frac{2}{3} - (2\delta_{\mathrm{kh}} + S_{\max})/D > 0$, and $m \le 3\eta_1 D$ (H14)–(H15). Then on $\mathcal{U}_{\mathrm{post}} = \{(\sigma, \psi, w_{\mathrm{post}}) : p < h_K\}$:*

(i) Well-definedness and C^7 seams. After $\mathcal{B}$, $p_2 < h_K + \delta_{\mathrm{kh}}$. On every sector piece, $X_0, X_1, X_2 \ge 0$ with $X_0 + X_1 + X_2 = 1$, and only the coordinate not carrying an angle can vanish (at $p = 0$, the angle matches polar coordinates $w_{\mathrm{post}} = \sqrt{p/\pi}e^{2\pi i\vartheta}$ smoothly). On band j, for $|c| \le d_2$ we have $\tau_B(c) = 1$, so $p_2 > x_{0,B} > 0$ and $v_\ell \in (\theta_{0,B}, 1 - \theta_{0,B})$ avoids the branch cut $\{0, 1\}$; for $d_2 \le |c| < m$, $q(c) \equiv 0$, so $S(c) = |c|$ ($\lambda'(c) = 0$ for $c \ge d_2$, $\lambda'(c) = 1$ for $c \le -d_2$), making $e^{2\pi i T_a}$ and $e^{2\pi i(v_\ell - T_a)}$ independent of the integer lift of v_ℓ and smooth across $p_2 = 0$. Across all six seams $c_j = \pm m$ (where $\tau_B = 0$, $k_j(-m) = 0$, $k_j(+m) = \kappa_j$) and across the fundamental cut $v_c = -\frac{1}{6} \leftrightarrow \frac{5}{6}$ (via $\mathrm{off}_{2c} - \mathrm{off}_{2s} = (0, -\frac{1}{4})$ and τ_v), the rank-one projector $ZZ^* \in \mathbb{R}^9$ matches identically to order C^7 and is invariant under τ_u, τ_v.

(ii) Exact zero set of Z_0 (line avoidance). For $x \in \mathcal{U}_{\mathrm{post}}$, $Z_0(\mathcal{W}(\mathcal{B}(x))) = 0$ if and only if x lies in the (τ_u, τ_v)-orbit of one of the three straight marked fibres $(0, 0)$, $(0, \frac{1}{2})$, $(\frac{1}{2}, -\frac{1}{6})$.

(iii) Symplectic embedding and cancellation-free inverse. On $\mathcal{U}_{\mathrm{post}} \setminus \mathrm{deck}\{\mathcal{M}_1, \mathcal{M}_2, \mathcal{M}_3\}$, $K_{\mathrm{post}} = \chi_\kappa \circ \mathcal{W} \circ \mathcal{B}$ takes values in $B^\circ(1)$, satisfies $K_{\mathrm{post}}^*\omega_0 = d\sigma \wedge d\psi + dw_{\mathrm{post},x} \wedge dw_{\mathrm{post},y}$, and identifies two points if and only if they lie in the same (τ_u, τ_v)-orbit. The inverse algorithm in §7.3 recovers every sector moment directly from the non-negative coordinate

$p = X_2, X_0, X_1$ (never by $1 - X_a - X_b$) and every band moment from $p_2 = X_b - \lambda_-(c)$ ($c \geq 0$) or $X_a - \lambda_+(c)$ ($c < 0$).

Proof. (i) On band j, $x_r = 1 - 2p_2 - S(c) > 1 - 2(h_K + \delta_{\mathrm{kh}}) - S_{\max} = \frac{1}{3} + \frac{2}{3}D - 2\delta_{\mathrm{kh}} - S_{\max} = \frac{1}{3} + \eta_1 D > \frac{1}{3}$ by (H14). On each sector, $\min_l |c_l| \geq m > 0$ ensures both non-p coordinates are $\geq m > 0$ (for instance on sector 1, $c_1 \geq m \implies X_2 = p + c_1 \geq m > 0$ and $c_2 \leq -m \implies X_1 = -c_2 + p \geq m > 0$). At $c_0 = -m$: $k_0(-m) = 0$, $\mathrm{left}_0 = \mathrm{off}_{2c}$, $\lambda(-m) = -m$, and $\lambda'(-m) = 1$, giving $(x_r, x_a, x_b) = (1 - 2p + m, p, p + m)$ and $(T_a, v_\ell - T_a) = (\tilde{\vartheta} + \tilde{\psi}, -\tilde{\psi})$, matching sector $2c$; at $c_0 = +m$: $k_0(+m) = \kappa_0 = -\frac{1}{12}$, $\mathrm{left}_0 + (\kappa_0, 0) = (0, 0) = \mathrm{off}_0$, $\lambda(+m) = 0$, and $\lambda'(+m) = 0$, giving $(1-2p-m, p+m, p)$ and $(\tilde{\psi}, \tilde{\vartheta} - \tilde{\psi})$, matching sector 0. Similarly $c_1 = \pm m$ matches sectors $0 \leftrightarrow 1$ ($\mathrm{left}_1 + (\kappa_1, 3\kappa_1) = \mathrm{off}_1$) and $c_2 = \pm m$ matches sectors $1 \leftrightarrow 2s$ ($\mathrm{left}_2 + (0, 0) = \mathrm{off}_{2s}$). Under τ_v, a point near $v_c = -\frac{1}{6}$ in sector $2c$ maps to $v_c' = v_c + 1 \in [\frac{5}{6}, 1)$, where substituting $\sigma' = \sigma + 3 - 9p$ and $\vartheta' = \vartheta + 9\psi + \frac{1}{2}$ into sector $2s$'s formula gives $(X_0', X_1', X_2') = (X_0, X_1, X_2)$, $T_2' = T_2$, and $T_1' = (\vartheta + 9\psi + \frac{1}{2} + \frac{1}{4}) - 8(\psi + \frac{1}{12}) = \vartheta + \psi + \frac{1}{12} = T_1 \pmod 1$.

(ii) Since $|\kappa| = 1$, $Z_0 = 0 \iff W_2 = \kappa W_1$, which requires $X_1 = X_2$ and (when $X_1 = X_2 > 0$) $T_2 - T_1 \equiv -\frac{1}{12} \pmod 1$. Across all seven pieces:

- On band 1 ($X_1 = x_r, X_2 = x_a$), $X_1 - X_2 = 1 - 3p_2 - (\frac{3}{2}S(c) + \frac{1}{2}c) > 0.70D - (\frac{3}{2} \cdot 0.32 + 0.16)D = +0.06D > 0$; symmetrically $X_2 - X_1 > +0.06D > 0$ on band 2.
- On sectors 0, $2c$, $2s$: $X_1 - X_2 = \sigma \geq m > 0$ ($s = 0$), $X_1 - X_2 = \sigma \leq -m < 0$ ($s = 2c$), and $X_2 - X_1 = 3 - 9p - \sigma > \frac{1}{6}(3 - 9p) > 0$ ($s = 2s$).
- On band 0 ($X_1 = x_a, X_2 = x_b$): $X_1 = X_2 \iff \lambda_+(c) = \lambda_-(c) \iff c = 0$ ($v_c = 0$). At $c = 0$, $\tau_B(0) = 1$, $\tau_B'(0) = 0$, $\lambda'(0) = \frac{1}{2}$, $k_0(0) = -\frac{1}{24}$, and $\mathrm{left}_0 = (\frac{1}{12}, 0)$, so $T_a = \psi + \frac{1}{24} + \frac{1}{2}v_\ell$ and $T_2 - T_1 = v_\ell - 2T_a = -2\psi - \frac{1}{12}$ (v_ℓ cancels identically), which equals $-\frac{1}{12} \pmod 1$ if and only if $\psi \in \{0, \frac{1}{2}\} \pmod 1$ ($\mathcal{M}_1$ and $\mathcal{M}_2$).
- On sector 1: $X_1 = X_2 \iff 2 - 5p - \sigma = 4p + \sigma - 1 \iff v_c = \frac{1}{2}$ (where $|c_1| = |c_2| = \frac{1}{2}(1 - 3p) > \frac{1}{2}D > m$, outside all bands). There $X_1 = X_2 = \frac{1}{2}(1 - p) > \frac{1}{3}$ and $T_2 - T_1 = \tilde{\psi} = \psi + \frac{1}{12} \equiv -\frac{1}{12} \pmod 1 \iff \psi \equiv -\frac{1}{6} \pmod 1$ ($\mathcal{M}_3$).

(iii) On each sector, $(X_1, X_2) = A_s(p, \sigma) + B_s$ with $A_s \in \mathrm{GL}_2(\mathbb{Z})$ and $(T_1, T_2) = (\tilde{\vartheta}, \tilde{\psi})A_s^{-1}$ gives $\sum_{l=1}^{2} dX_l \wedge dT_l = dp \wedge d\vartheta + d\sigma \wedge d\psi$; on band j, $dx_a \wedge dT_a + dx_b \wedge d(v_\ell - T_a) = dc \wedge d\psi_2 + dp_2 \wedge dv_\ell = d\sigma_{\mathrm{out}} \wedge d\psi_{\mathrm{out}} + dp_2 \wedge d\vartheta_{\mathrm{out}}$, and $\mathcal{B}$ is symplectic by Proposition 3.2. Pulling back ω_0 by χ_κ gives ω_{FS}, whose pullback by W is $\sum_{l=1}^{2} dX_l \wedge dT_l$. Finally, consider the three band tests $\mathcal{R}_j = \{|X_{j+1} - X_{j+2}| < m,\ X_j > \frac{1}{3}\}$ (indices modulo 3). Although the sets $\mathcal{R}_j$ overlap abstractly on the simplex Δ^2, every actual image point of $\mathcal{W} \circ \mathcal{B}(\mathcal{U}_{\mathrm{post}})$ satisfies $p_2 < h_K + \delta_{\mathrm{kh}} = \frac{1}{3} - \frac{1}{3}(D - 3\delta_{\mathrm{kh}}) = \frac{1}{3} - \frac{7}{30}D$, and on this image each band j satisfies $\mathcal{R}_j$ exclusively while no sector point satisfies any $\mathcal{R}_l$: on band 0, $X_0 = x_r > \frac{1}{3}$ and $|X_1 - X_2| = |c| < m$ put the point in $\mathcal{R}_0$; if also $X_1 = p_2 + \lambda_+(c) > \frac{1}{3}$, then since $\max_{|t| \leq d_2} \lambda_+(t) = 0.20D < \frac{7}{30}D$ we must have $c > d_2$, so $S(c) = c$, $\lambda_-(c) = 0$, and $X_0 - X_2 = 1 - 3p_2 - c > \frac{7}{10}D - 0.32D = 0.38D > m$ (excluding $\mathcal{R}_1$); symmetrically $X_0 - X_1 > 0.38D > m$ whenever $X_2 > \frac{1}{3}$ (excluding $\mathcal{R}_2$). Conversely, on sector 0 ($0 \leq v_c < \frac{1}{3}$, $\sigma \geq m$, $1 - 3p - \sigma \geq m$, $p < h_K < \frac{1}{3}$), $X_1 - X_2 = \sigma \geq m$ excludes $\mathcal{R}_0$, $X_0 - X_2 = 1 - 3p - \sigma \geq m$ excludes $\mathcal{R}_1$, and $X_2 = p < \frac{1}{3}$ excludes $\mathcal{R}_2$; sectors 1, $2s$, and $2c$ are analogous. Outside the band images, $\mathrm{argmin}(X_0, X_1, X_2)$ uniquely identifies sector 0 ($X_2 = p$ minimal), sector 1 ($X_0 = p$ minimal), or sector $2s/2c$ ($X_1 = p$ minimal: on $2c$, $X_2 \geq p + m > p$ and $X_0 - X_1 = 1 - 3p + \sigma \geq \frac{1}{2}(1 - 3p) > 0$; since $2c$ is the τ_v^{-1}-image of $v_c' \in [\frac{5}{6}, 1)$ where $2s$'s formula produces the identical W, $2s$'s inverse followed by `reduce` recovers $(v_c, \psi, p, \vartheta)$). □

7. Explicit parameter verification and proof of Theorem 1.1

7.1. The closed-form parameter tuple $\Pi^*(k, \varepsilon_c; \frac{1}{4})$.

Definition 7.1 (Parameter tuple $\Pi^*(k,\varepsilon_c;f)$)**.** For an integer $k\ge 10$, $\varepsilon_c\in(0,\frac{1}{100}]$, and $f=\frac14$, put $s=\sqrt{k}$, $\eta=\varepsilon_c/(1+\sqrt{1-\varepsilon_c})$, $t_0=1-\frac12\eta$, $a=t_0/s$, $h=t_0/3$, $c_s=3s-9$, $g=\frac32\eta$, $T=g/c_s$, $(\lambda_a,\lambda_2,\mu)=(\frac{9}{10},\frac{3}{10},\frac12)$, $w_G=\lambda_a g$, $w_2=\lambda_2 g$, $\upsilon=\sqrt{11/1000}$.

(1) *Comparator keyhole K_C:* $r_a=\sqrt{a/\pi}$, $d=\sqrt{fT/\pi}/(1+\upsilon)$, $\beta=fT/\big(\pi(r_a+d)(r_a+8.4d)\big)$, $W_1=\beta/(3(1+\beta))$, $r_h=0.9d\sin(2\pi W_1)$, $\rho_1=0.8((d-r_h)/(d+r_a))^2$, $W_2=\beta/(1+\beta-\rho_1)-W_1$, $X_w=0.2\pi(d-r_h)^2$.

(2) *Post height h_K and $D=1-3h_K$:* $h_K=\max\big(a,\ O(-w_G)-\frac12\mu T(1-\lambda_a),\ h-\frac12\mu T\big)$, so $D=\min\big(\frac{s-3+\frac32\eta}{s},\ \frac{s-3+w_G}{s}+\frac32\mu T(1-\lambda_a),\ \frac12\eta+\frac32\mu T\big)>0$.

(3) *Corner keyhole K_B:* $(m,d_2,\sigma_0,\delta_{\mathrm{kh}})=(0.32,0.20,0.20,0.10)D$, $r_B=\sqrt{h_K/\pi}$, $d_B=\sqrt{0.8\delta_{\mathrm{kh}}/\pi}/1.05$, $\beta_B=0.8\delta_{\mathrm{kh}}/\big(\pi(r_B+d_B)(r_B+8.4d_B)\big)$, with $(W_{1,B},r_{h,B},\rho_{1,B},W_{2,B},X_{w,B})$ by the same 'keyhole_star' formulas with ρ_1-factor 0.7.

7.2. **Parameter hypotheses and their uses.** The labels used in the parameter verification below have the following mathematical meaning. For either keyhole, its positive parameters satisfy $0<W_1<W_2\le 1/6$, $0<\rho_1<1$ and the exact relations (2); its budgets $\delta_{\mathrm{rel}},\delta_{\mathrm{small}}$ are those of Theorem 3.3. Subscript B denotes the corner keyhole.

(H0) : $k\in\mathbb{Z}$, $k\ge 10$, $0<\varepsilon_c\le 1/100$ (the parameter regime).
(H1) : $0<\lambda_2<\lambda_a<1$ and $0<\mu<1$ (the comparator schedules).
(H2) : $0<\rho_b<a$, equivalently $r_+>1$, where $\rho_b=R_s^{-2}$ and $a=r_+^2\rho_b$ (the enlarged source domain).
(H3)–(H5): K_C has input radius $r_a=\sqrt{a/\pi}$ and satisfies the positive-parameter conditions and (2) (Theorem 3.3 for the comparator).
(H6)–(H7): The two schedule inequalities displayed in Theorem 4.2 hold for every $u\in[0,1]$ (stack containment).
(H8) : $\pi d^2\le T(1-\mu-\lambda_2/2)$ and $\pi d^2+\lambda_2 T/2+\mu T\le O(-w_G)$ (the remaining comparator branches).
(H9) : $c_s\pi r_h^2+\frac32\lambda_2 g\le g$. This is an additional implementation guard, also verified below; the proof of Theorem 4.2 uses (H6)–(H8) directly.
(H9b) : $3-9a>\delta$, where $\delta=w_2/4$ (separation of the group windows from the marked-fibre spirals).
(H10) : The constants and integer shifts are those of Construction 5.1; its four v-windows are pairwise disjoint, and its two u-lobes are disjoint and miss 1/3 and 1 as specified in Proposition 5.2(ii) (marked-fibre routing).
(H11) : $a\le h_K<1/3$, $D=1-3h_K$, and $h_K>\max_{G\in[-w_G,G_{\max}]}(O(G)-\mu\,\mathrm{slack}(G))$ (the common post-chart domain in (9)).
(H12) : K_B has input radius $r_B=\sqrt{h_K/\pi}$ and satisfies the positive-parameter conditions and (2) (the corner keyhole).
(H13) : $\max(\pi d_B^2,\delta_{\mathrm{rel},B},\delta_{\mathrm{small},B})\le\delta_{\mathrm{kh}}$ (post-chart fibre containment).
(H14) : $0<d_2<m$ and $\eta_1=\frac23-(2\delta_{\mathrm{kh}}+S_{\max})/D>0$, where $S_{\max}=\max_{|c|\le m}\sqrt{c^2+q(c)^2}$ (positive simplex coordinates).
(H15) : $m\le 3\eta_1 D$ (separation in the simplex transfer).
Split : $2m<D-3\delta_{\mathrm{kh}}$ (disjoint post-chart bands before and after the corner keyhole).

Thus "(H0)–(H15)" groups the parameter conditions; the split inequality is included whenever all transfer hypotheses are invoked. The program checks these conditions by margin guards, with tolerances for rounded versions of exact identities such as (K1). The guards' count and floating-point tolerances are implementation details, not additional mathematical hypotheses.

Lemma 7.2 (Grid-free analytic bounds for (H6) and (H7))**.** *With* $(\lambda_a, \lambda_2, \mu, f) = (\frac{9}{10}, \frac{3}{10}, \frac{1}{2}, \frac{1}{4})$, *suppose* $\delta_{\mathrm{rel}} = fT = \frac{1}{4}T$ *and* $\delta_{\mathrm{small}} \le 0.04T$. *Then (H6) holds on all of* $[0,1]$ *with margin* $\ge \min(0.05T - \delta_{\mathrm{small}}, 0.0128T) \ge 0.01T$ *(and* $> 0.0128T$ *whenever* $\delta_{\mathrm{small}} \le 0.037T$, *as for* Π^* *below where* $\delta_{\mathrm{small}} < 0.007T$*), and (H7) holds on all of* $[0,1]$ *with margin* $\ge 0.10T$.

Proof. Divide by T. For (H6), $(1-\mu)(1-\lambda_a+u(\lambda_a-\lambda_2)) = 0.05+0.30u \ge 0.05 > 0.04 \ge \delta_{\mathrm{small}}/T$. It remains to bound $h_6(u) = 0.05 + 0.30u - 0.25S_4(u)$ on $[0,1]$. On $[0, \frac{1}{2}]$, Lemma 2.1(i) gives $S_4(u) \le u$, so $h_6(u) \ge 0.05+0.05u \ge 0.05$. On $[\frac{1}{2}, 1]$, S_4 is concave, so for $u_0 = \frac{2}{3}$ ($S_4(\frac{2}{3}) < 0.9414$, $S_4'(\frac{2}{3}) = \frac{256}{70}\sin^8(\frac{2\pi}{3}) = \frac{81}{70} < 1.1572$), $S_4(u) \le S_4(u_0) + S_4'(u_0)(u-u_0)$, bounding $h_6(u)$ from below by an affine function whose values at $u = \frac{1}{2}$ and $u = 1$ are > 0.0128 and > 0.0182. For (H7), since $\max(\delta_{\mathrm{rel}}, \delta_{\mathrm{small}})/T = \frac{1}{4}$, we must bound $h_7(u) = 0.5(0.7 + 0.3u) - 0.25 - 0.3I_4(u) = 0.10 + 0.15u - 0.30I_4(u)$ on $[0,1]$. Its derivative $0.30(\frac{1}{2} - S_4(u))$ is ≥ 0 on $[0, \frac{1}{2}]$ and ≤ 0 on $[\frac{1}{2}, 1]$, so $\min_{[0,1]} h_7 = \min(h_7(0), h_7(1)) = 0.10 > 0$. □

Proposition 7.3 (Universal validity of (H0)–(H15))**.** *For every integer* $k \ge 10$ *and every* $\varepsilon_c \in (0, \frac{1}{100}]$, $\Pi^*(k, \varepsilon_c; \frac{1}{4})$ *satisfies all hypotheses of Theorems 3.3, 4.2, 5.2, and 6.1, as well as* $r_+ > 1$.

Proof. Since $s \ge \sqrt{10} > 3.16227$ and $0 < \eta \le 1 - \sqrt{0.99} < 0.005013$, we have $t_0 > 0.9974$, $s/(s-3) \le \sqrt{10}/(\sqrt{10}-3) = 10 + 3\sqrt{10} < 19.49$, and

$$\frac{T}{a} = \frac{\frac{1}{2}\eta/(s-3)}{t_0/s} = \frac{\eta}{2t_0}\,\frac{s}{s-3} < \frac{0.005013}{1.9948} \cdot 19.49 < 0.049.$$

The parameter ranges (H0)–(H1) hold by definition and the fixed values $(\lambda_a, \lambda_2, \mu) = (9/10, 3/10, 1/2)$. The radius identities, integer shifts, and $D = 1 - 3h_K$ are exact definitional identities; positivity of the remaining keyhole parameters follows from their displayed formulas.

(H2) and $r_+ > 1$: Since $0 < m_s \le \frac{1}{8}\eta$ and $\eta < 0.01$, $(1-m_s)^2 > 1 - 2m_s \ge 1 - \frac{1}{4}\eta$, while $(1-\frac{1}{2}\eta)/(1-\eta) = 1 + \frac{\eta}{2(1-\eta)} > 1 + \frac{1}{2}\eta$. Thus $r_+^2 = (1-m_s)^2(1-\frac{1}{2}\eta)/(1-\eta) > (1-\frac{1}{4}\eta)(1+\frac{1}{2}\eta) = 1 + \frac{1}{4}\eta(1-\frac{1}{2}\eta) > 1$, which is equivalent to $\rho_b = R_s^{-2} = (1-\eta)/(\sqrt{k}(1-m_s)^2) < a$.

(H3)–(H5) for K_C: Since $\pi(r_a + d)(r_a + 8.4d) > \pi r_a^2 = a$, $\beta < \frac{1}{4}(T/a) < 0.0123$. Also $\pi d^2 = \frac{1}{4}T/(1+\upsilon)^2 < 0.205T$, so $(d/r_a)^2 < 0.205(0.049) < 0.0101$ ($d/r_a < 0.101$). By definition, (K1) and (K2) hold with equality/$0.9 < 1$, $r_h < 0.9d(2\pi W_1) < 0.024d < d$, $\rho_1 < 0.8(0.101/1.101)^2 < 0.0068 < 1$, $W_2 - W_1 = \beta\big(\frac{1}{1+\beta-\rho_1} - \frac{2}{3(1+\beta)}\big) \ge \frac{\beta}{3(1+\beta)} > 0$, and $W_2 < W_1 + W_2 = \beta/(1+\beta-\rho_1) < 0.0123/0.993 < 0.013 < \frac{1}{6}$. For (K3), $\rho_1\pi(d+r_a)^2 + \frac{1}{2}X_w = (0.8+0.1)\pi(d-r_h)^2 = 0.9\pi(d-r_h)^2 < \pi(d-r_h)^2$ holds identically! Moreover, $d/(r_a + 8.4d) < \frac{1}{8.4}$ and $W_2/\beta < \frac{1}{0.993} - \frac{1}{3.04} < 0.68$, so

$$\frac{\delta_{\mathrm{small}}}{fT} = 4\pi^2\,\frac{d}{r_a + 8.4d}\Big(\frac{W_2}{\beta}\Big)^2\beta < \frac{4\pi^2}{8.4} \cdot (0.68)^2 \cdot 0.0123 < 0.027 < 0.04.$$

Lemma 7.2 then gives (H6) and (H7).

(H8), (H9), (H9b): $\pi d^2 < 0.205T < 0.35T = T(1-\mu-\frac{1}{2}\lambda_2)$, and $s(\pi d^2 + 0.65T) < 0.855sT = 0.4275\eta\frac{s}{s-3} < 0.042 < 1 - 0.45\eta = sO(-w_G)$, giving (H8). For (H9), $c_s\pi r_h^2 < c_s\pi(0.024d)^2 < 6\cdot10^{-4}c_s(0.205T) < 2\cdot10^{-4}g$, so $c_s\pi r_h^2 + 1.5\lambda_2 g < (0.45+0.001)g < g$. For (H9b), $a < 1/\sqrt{10} \implies 3 - 9a > 3 - 9/\sqrt{10} = 3(1 - 3/\sqrt{10}) > 0.1539$, while $\delta = \frac{1}{4}w_2 = \frac{9}{80}\eta < 0.0006 < 0.1539$. (H10) is a check of fixed rational constants ($0.045 + 0.045 < 0.10$, $0.045 + 0.14 < 0.25$, $2 \cdot 0.12 < \frac{1}{4}$, $\frac{1}{6} - 0.12 - 0.02 > 0.02 > 0$).

(H11)–(H15): By Definition 7.1(2), $D > 0$ ($h_K < \frac{1}{3}$), $h_K \ge a$, and since $O(G) - \mu\,\mathrm{slack}(G)$ is piecewise affine on $[-w_G, 0] \cup [0, G_{\max}]$, its maximum is bounded by $\max\big(O(-w_G) - \mu T(1-\lambda_a), O(0) - \mu T, O(G_{\max}) - \mu T\big)$; note $O(0) < O(G_{\max}) = h$ when $G_{\max} > 0$ ($O(-w_G)$ otherwise), and subtracting $\frac{1}{2}\mu T(1-\lambda_a) > 0$ in h_K ensures strict inequality $h_K > O(G) - \mu\,\mathrm{slack}(G)$

everywhere on $[-w_G, G_{\max}]$. Finally, since $3s-9 \geq 3\sqrt{10}-9 > 0.48683$ and $\eta < 0.005013$, we have $\frac{3}{2}\mu T = \frac{9\eta}{8(3s-9)} < 0.01159$, so $D \leq \frac{1}{2}\eta + \frac{3}{2}\mu T < 0.01410 < 0.015$, $\delta_{\mathrm{kh}} = 0.10D < 0.0015$, and $h_K = \frac{1}{3}(1-D) > 0.32833$. Setting $x = d_B/r_B = \frac{1}{1.05}\sqrt{0.8\delta_{\mathrm{kh}}/h_K} < 0.058$ and retaining the full denominator of Definition 7.1(3), the rational function $t \mapsto 1.05^2 t^2/\big((1+t)(1+8.4t)\big)$ is strictly increasing on $(0,\infty)$, so

$$\beta_B = \frac{0.8\delta_{\mathrm{kh}}}{\pi(r_B+d_B)(r_B+8.4d_B)} = \frac{1.05^2 x^2}{(1+x)(1+8.4x)} < 0.0024 < 0.003.$$

Hence $\rho_{1,B} < 0.7\big(\frac{x}{1+x}\big)^2 < 0.0022 < 1$, $W_{2,B}/\beta_B = \frac{1}{1+\beta_B-\rho_{1,B}} - \frac{1}{3(1+\beta_B)} < \frac{1}{1-0.0022} - \frac{1}{3(1.0024)} < 0.68$, and $W_{2,B} < 0.68\beta_B < 0.0017 < \frac{1}{6}$. Exactly as for K_C, K_B satisfies (K1)–(K3) with $\rho_{1,B}\pi(d_B+r_B)^2 + \frac{1}{2}X_{w,B} = (0.7+0.1)\pi(d_B-r_{h,B})^2 = 0.8\pi(d_B-r_{h,B})^2$, $\delta_{\mathrm{rel},B} = 0.8\delta_{\mathrm{kh}}$, $\pi d_B^2 = 0.8\delta_{\mathrm{kh}}/1.05^2 < \delta_{\mathrm{kh}}$, and

$$\frac{\delta_{\mathrm{small},B}}{\delta_{\mathrm{kh}}} = 0.8\cdot 4\pi^2 \frac{d_B}{r_B+8.4d_B}\Big(\frac{W_{2,B}}{\beta_B}\Big)^2\beta_B < 0.8\cdot\frac{4\pi^2}{8.4}\cdot(0.68)^2\cdot 0.0024 < 0.0042 < 0.01,$$

proving (H12)–(H13). On $[-m,m]$, for $|c| \leq d_2$ we have $S(c) \leq \sqrt{d_2^2+\sigma_0^2} = 0.20\sqrt{2}\,D < 0.283D < 0.32D = m$, and for $d_2 \leq |c| \leq m$, $S(c) = |c| \leq m$; thus $S_{\max} = 0.32D$, $\eta_1 = \frac{2}{3} - (0.20+0.32) = \frac{11}{75} > 0.1466 > 0$, $3\eta_1 = \frac{11}{25} = 0.44 \geq 0.32 = m/D$ (H14–H15), and $2m/D + 3\delta_{\mathrm{kh}}/D = 0.64 + 0.30 = 0.94 < 1$ ('split'). □

7.3. **Exact left inverse, non-membership certification, and proof of Theorem 1.1.** The exact-real inverse algorithm. We now define Preimage(i,Y) mathematically, including its conventions away from the forward image. Failure is denoted by $\perp$ (or by ok = false). Only operations on the selected branch are evaluated; an undefined real operation returns $\perp$. Write $\mathrm{frac}(t) = t - \lfloor t \rfloor$, $\mathrm{cart}(p,\theta) = \sqrt{p/\pi}\,e^{2\pi i\theta}$ for $p \geq 0$, and set $\arg 0 = 0$ when a polar angle at the origin is needed. All scalar inverse equations use their unique roots from Sections 2.2–3.

1. *Undo the ball chart.* If $|Y| > R_s$, return $\perp$. Otherwise put $y = Y/(\sqrt{\pi}R_s)$ and $\zeta = \sqrt{\pi}(y_1+iy_3, y_2+iy_4)$, and form $Z = (\sqrt{1-|\zeta|^2}, \zeta_1, \zeta_2) \in S^5$. Undo the unitary rotation of Section 6.1 to obtain W. Set $X_j = |W_j|^2$ and $t_j = \arg W_j/(2\pi)$.
2. *Undo the simplex chart.* Test $\mathcal{R}_j = \{|X_{j+1} - X_{j+2}| < m,\ X_j > 1/3\}$ for $j = 0,1,2$ (indices modulo 3). If more than one test succeeds, use the largest j, matching the last-write convention of the implementation. On this branch let $a = j+1$, $b = j+2$, $c = X_a - X_b$, and
$$p_2 = \max\left(0, \begin{cases} X_b - \lambda_-(c), & c \geq 0, \\ X_a - \lambda_+(c), & c < 0. \end{cases}\right).$$
Put $A = t_a - t_j$, $B = t_b - t_j$, $v_\ell = \mathrm{frac}(A+B)$, $\psi_2 = A - \lambda'(c)v_\ell$, and, with $(a_j, b_j) = \mathrm{left}_j$,
$$\sigma = c + j(1-3p_2), \quad \psi = \psi_2 - k_j(c) - a_j, \quad \vartheta = v_\ell + 3j\psi_2 - 3jk_j(c) - b_j.$$
If no $\mathcal{R}_j$ succeeds, choose the least index attaining $\min(X_0, X_1, X_2)$. With $T_1 = t_1 - t_0$, $T_2 = t_2 - t_0$, the inverse sector coordinates before the offsets are

minimal index	p_2	σ	$\widetilde{\psi}$	$\widetilde{\vartheta}$
2	X_2	$X_1 - X_2$	T_1	$T_1 + T_2$
0	X_0	$1 + X_2 - 4X_0$	$-T_1 + T_2$	$-5T_1 + 4T_2$
1	X_1	$3 - X_2 - 8X_1$	$-T_2$	$T_1 - 8T_2$

Subtract off_0, off_1, or off_{2s} respectively from $(\widetilde{\psi}, \widetilde{\vartheta})$. In both cases reduce by τ_v^{-n} with $n = \lfloor \sigma/(3-9p_2) + 1/6 \rfloor$, and set $z = (\sigma, \psi, \mathrm{cart}(p_2, \vartheta))$ using the reduced coordinates.

3. *Undo the corner keyhole and post chart.* Define the branchwise inverse $\mathcal{B}_-$ by the same reduction and band tests $|c_j| < m$ as $\mathcal{B}$. In each active band replace $K_B^{\tau_B}$ by its global inverse and replace the conjugate drift $-\tau_B' g_B$ by $+\tau_B' g_B$; preserve c_j, restore the deck translate, and keep the last active band if several tests succeed. Outside all bands use the identity. Put $x = \mathcal{B}_-(z)$. For $x = (\sigma, \psi, \mathrm{cart}(p, \vartheta))$, set

$$u_a = -\tfrac{1}{6} - \psi, \quad v = \frac{\sigma}{P(p)}, \quad G = (u_a - 1)P(p), \quad w = \mathrm{cart}(p, \vartheta + 9(u_a - 1)v).$$

Undo Φ_g and Ψ_S, then apply T_u^{-n} with $n = \lfloor (G - J(p))/P(p) \rfloor$. This subtracts $nP(p)$ from G and rotates w by $e^{-2\pi i 9nv}$.

4. *Undo the comparator and cell map.* For $i < 9$ the comparator is the identity. For $i \geq 9$ undo F_2 first: where $c(G) > 0$, for the input fibre $w_2 = \mathrm{cart}(x_2, \theta_2)$, set

$$w_1 = \mathrm{cart}(\max(x_2 - c(G), 0), \theta_2), \qquad v_1 = v_2 + c'(G)\big(\theta_2 - \tfrac{1}{2}\operatorname{sgn}\theta_2\big),$$

using the principal angle in $(-1/2, 1/2]$; where $c(G) = 0$ use the identity. Then undo F_1 by the global inverse keyhole and its opposite conjugate drift. Denote the resulting flat point by (G_0, v_0, w_0), with $p_0 = \pi|w_0|^2$. For $i < 9$ set

$$T_c = 9v_0 - (i-4) - 9\left\lfloor \frac{9v_0 - (i-4)}{9} \right\rfloor, \quad X_c = (G_0 - J(p_0))/9, \quad w_c = w_0 e^{2\pi i s T_c/3}.$$

For $i \geq 9$ set

$$T_c = \mathrm{frac}(v_0 - \tfrac{1}{2}), \quad X_c = G_0 - J(p_0) - i(a - p_0), \quad w_c = w_0 e^{2\pi i(3s-i)T_c}.$$

5. *Accept or reject.* Require $p_0 < a$, $0 < X_c < a - p_0$, and $0 < T_c < 1$; otherwise return $\perp$. Apply the unique inverse KR^{-1} on $\mathcal{C}_a$, undo S_{in}, and return (X, true). The test for membership in the closed unit-ball image is then $|X| \leq 1$.

This is an exact-real algorithm for the stage formulas, with explicit failure conventions. Fixed iteration counts, floating-point guards and their error bounds are separate from this definition. Code names such as `to_fund`, `flat_to_cell` and `invert` implement its reductions and branch rules.

Lemma 7.4 (Forward-reconstruction soundness of Preimage)**.** *For any $Y \in \mathbb{R}^4$ and any $0 \leq j < k$:*

(i) If $Y = f_j(X)$ with $X \in \mathbb{B}^\circ(r_+)$, then $\mathrm{Preimage}(j, Y) = (X, \mathrm{true})$.

(ii) Conversely, if $\mathrm{Preimage}(j, Y) = (X, \mathrm{true})$ *with $|X| < r_+$ (in particular if $|X| \leq 1$), then $Y = f_j(X)$; hence* $\mathrm{Preimage}(j, Y)$ *returns $|X| \leq 1$ and* $\mathrm{ok} = \mathrm{true}$ *if and only if $Y \in f_j(\mathbb{B}(1))$.*

Proof. The first step rejects $|Y| > R_s$. Otherwise it constructs $W \in S^5$ with rotated coordinate $Z_0 = \sqrt{1 - |Y|^2/R_s^2}$. When $|Y| < R_s$, the point $y = Y/(\sqrt{\pi}R_s) \in B^\circ(1)$ therefore satisfies $\chi_\kappa([Z(W)]) = y$; when $|Y| = R_s$, the point lies on L_κ.

(i) Suppose $Y = f_j(X)$ with $X \in \mathbb{B}^\circ(r_+)$. By Theorem 6.1(iii), $\mathcal{B}^{-1} \circ (\chi_\kappa \circ \mathcal{W})^{-1}(y)$ lies in the (τ_u, τ_v)-orbit of $\Pi_{\mathrm{post}}(\Psi(y_{\mathrm{fl}}))$, where $y_{\mathrm{fl}} = (F^{(j)} \circ C_j \circ \mathrm{KR} \circ S_{\mathrm{in}})(X)$ satisfies $J(p) < G < K(p)$ by (8) and (10). Applying $\Psi^{-1} \circ \Pi_{\mathrm{post}}^{-1}$ gives a point in the $\langle T_u, T_v \rangle$-orbit of y_{fl}. The T_u-reduction brings G into the unique representative $J(p) \leq G' < K(p)$, which therefore equals $T_v^m(y_{\mathrm{fl}}) = (G, v + m, w)$ for some $m \in \mathbb{Z}$. Next, $(F^{(j)})^{-1}$ commutes with T_v (shifting only v by $+m$), so $(F^{(j)})^{-1}(G, v + m, w) = (G_0, v_0 + m, w_0)$ where $(G_0, v_0, w_0) = C_j(\mathrm{KR}(S_{\mathrm{in}}(X)))$ has $T_{\mathrm{cell}} \in (0, 1)$. In the cell step, for $j < 9$ we have $9(v_0 + m) - (j - 4) = T_{\mathrm{cell}} + 9m$, whose reduction modulo 9 into $[0, 9)$ recovers $T_{\mathrm{cell}} \in (0, 1)$ uniquely; for $j \geq 9$ we have $(v_0 + m) - \frac{1}{2} = T_{\mathrm{cell}} + m$,

whose reduction modulo 1 into $[0,1)$ recovers $T_{\mathrm{cell}} \in (0,1)$ uniquely. Thus the cell inverse recovers the original tuple $\mathrm{KR}(S_{\mathrm{in}}(X)) \in \mathcal{C}_a$, and $S_{\mathrm{in}}^{-1} \circ \mathrm{KR}^{-1}$ returns (X, true).

(ii) Suppose conversely that $\mathrm{Preimage}(j, Y) = (X, \mathrm{true})$ with $|X| < r_+$. The ball-chart step gives $|Y| \le R_s$ and a projective point $[W]$; the boundary case will be excluded at the end. We prove reconstruction in the order of the inverse algorithm, without assuming that $[W]$ belongs to a forward image.

Comparator reconstruction. The cell test and bijectivity of KR imply that forward evaluation from X recovers precisely the accepted cell tuple. The explicit formulas for C_j^{-1} and C_j then recover the flat point up to an integer T_v translate. For $j \ge 9$ the only possible loss of information before that cell test is the truncation $\max(x_2 - c(G), 0)$ in F_2^{-1}. If $c(G) > 0$, then $G > -w_2$ and $\tau(G) = 1$. Since the accepted fibre has $p < a$, Theorem 3.3(ii) puts its image under KS_1 strictly outside the closed hole $\{\pi|w_1|^2 \le x_0\}$ and the slit $|\theta_1| \le \theta_0$. A truncated inverse point has $w_1 = 0$, so its preimage under the global bijection KS_1 cannot have $p < a$. The same argument excludes a slit point. Hence neither information loss can occur on an accepted branch; F_1 and F_2 really cancel their inverses there. When $c(G) = 0$ there is no truncation. The twists are global bijections preserving p and commuting with the deck transformations. Restoring the explicit T_u and T_v translates therefore identifies the recovered post point $x = \mathcal{B}_-(z)$ with the post-chart image of the forward cell/comparator/twist chain, modulo (τ_u, τ_v). In particular

$$p(x) < h_K. \tag{13}$$

This uses only the stages before the transfer, not soundness of the transfer inverse.

Conditional inverse of the corner band. We claim that

$$\mathcal{B}_-(z) = x, \quad p(x) < h_K \quad \Longrightarrow \quad \mathcal{B}(x) = z, \quad p(z) < h_K + \delta_{\mathrm{kh}}.$$

If no band was active, both procedures are the identity. Otherwise take the last band l written by $\mathcal{B}_-$. Its c_l is preserved by the keyhole. At the returned source point put $L = 1 - 3p(x) > D$. Since $|c_l| < m = 8D/25$, we have $|c_l|/(3L) < 8/75 < 1/6$. Thus the source reduction uses exactly the same integer translate as the inverse procedure: the reduced coordinate is $c_l + lL$, with $l = 0, 1, 2$. Also $L > D > 2m$ makes this the unique active source band. The global planar keyhole and its inverse now cancel, as do their conjugate drifts; restoring the deck translate gives $\mathcal{B}(x) = z$. The full-disc nesting estimate of Theorem 3.3(i), with (H13), yields the second conclusion. Consequently the point produced by the simplex inverse satisfies

$$p_2 = p(z) < h_K + \delta_{\mathrm{kh}} = \frac{1}{3} - \frac{7}{30}D, \qquad L_2 = 1 - 3p_2 > \frac{7}{10}D > 2m. \tag{14}$$

Simplex band branches and their truncation. Suppose the simplex inverse selected $\mathcal{R}_l$. It sets $c = X_a - X_b \in (-m, m)$ and $\sigma = c + lL_2$ before reduction. Since $|c|/(3L_2) < 16/105 < 1/6$, this is already in the prescribed fundamental strip and belongs to band l alone. For $d_2 \le |c| < m$, $q(c) = 0$; the smaller of $\lambda_+(c), \lambda_-(c)$ is zero, so the recovered raw moment is a non-negative simplex coordinate and is not truncated. For $|c| < d_2$, $\tau_B(c) = 1$. If the raw moment had been negative and truncated to zero, z would have fibre $w_2 = 0$, contrary to Theorem 3.3(ii) for the source point in (13). The same hole and slit clearance validates its polar phase. Thus the recovered moment equals the raw one, and

$$p_2 + \lambda_+(c) = X_a, \qquad p_2 + \lambda_-(c) = X_b, \qquad 1 - 2p_2 - \lambda_+(c) - \lambda_-(c) = X_l.$$

The inverse formulas for ψ_2 and ϑ likewise recover the two relative phases of W. This proves reconstruction for the selected branch even if several $\mathcal{R}_j$ tests initially succeeded; no disjointness assumption on the whole simplex has been used. On an accepted input the reconstructed point is in the forward transfer domain, where Theorem 6.1(iii) gives exclusivity.

Simplex sector branches. If no $\mathcal{R}_j$ succeeded, the inverse takes $p_2 = \min_j X_j$. The affine moment inverses in the algorithm place σ in the corresponding sector interval, possibly as the deck-equivalent $2c$ representative. Suppose this recovered post point lay in a band, with $d = |c_l| < m$. In these sector formulas the two smaller simplex coordinates are p_2 and $p_2 + d$, so the remaining coordinate is

$$1 - 2p_2 - d > \frac{1}{3} + \left(\frac{7}{15} - \frac{8}{25}\right) D = \frac{1}{3} + \frac{11}{75} D > \frac{1}{3}.$$

The corresponding $\mathcal{R}_l$ would have succeeded, a contradiction. Hence the point is in the forward sector domain. The $\mathrm{GL}_2(\mathbb{Z})$ angle inverse and affine moment formulas recover $[W]$ there. The $2s/2c$ cut is covered by the deck identity in the proof of Theorem 6.1; at $|c| = m$ the sector and band formulas match, and at $|c| = d_2$ one has $q = 0$. A minimal-moment tie would put the point on an excluded band, so it cannot occur on an accepted sector input satisfying (14).

We have proved that the forward transfer from the accepted source reconstructs the original projective point $[W]$. If $|Y| = R_s$, its rotated representative has $Z_0 = 0$, contradicting forward line avoidance in Theorem 6.1(ii). Thus $|Y| < R_s$, where the ball chart and its inverse are genuine two-sided inverses. Undoing the scale gives $f_j(X) = Y$, as required. □

Proof of Theorem 1.1. Fix $k \geq 10$ and $\varepsilon \in (0, 1)$. Since $\varepsilon_c = \min(\varepsilon, \frac{1}{100}) \leq \varepsilon$, $R_c \leq R_{\mathrm{req}}$, and Proposition 7.3 gives $r_+ > 1$ and all hypotheses (H0)–(H15).

(1) Let $X = (q_1, q_2, p_1, p_2) \in \mathbb{B}^\circ(r_+)$. Stage 1, $S_{\mathrm{in}}(X) = (q_1, p_1, q_2, p_2)/(\sqrt{\pi} R_s)$, satisfies $\pi |S_{\mathrm{in}}(X)|^2 = |X|^2/R_s^2 < r_+^2 \rho_b = a$ and pulls back $dx \wedge dy + \omega_V$ to $(\pi R_s^2)^{-1} \omega_0$.
(2) By Proposition 2.2, KR maps $B^\circ(a)$ real-analytically, bijectively, and symplectically onto $\mathcal{C}_a$.
(3) By Lemma 4.1 and Theorem 4.2, $F^{(i)} \circ C_i$ $(0 \leq i < k)$ are C^7 symplectic embeddings on $\mathcal{C}_a$ whose images $U_i = F^{(i)}(C_i(\mathcal{C}_a))$ are pairwise disjoint modulo $\langle T_u, T_v \rangle$, lie in $\{p < h_K\}$, and miss the corridor $\mathcal{K}_{\mathrm{free}}$.
(4) By Proposition 5.2, $\Psi = \Phi_g \circ \Psi_S$ is a C^4 deck-equivariant symplectomorphism preserving $\{p < h_K\}$ and mapping $\Psi(U_i)$ into $\{p < h_K\} \setminus \mathrm{deck}\{\mathcal{M}_1, \mathcal{M}_2, \mathcal{M}_3\}$, keeping $\Psi(U_0), \ldots, \Psi(U_{k-1})$ pairwise disjoint modulo $\langle T_u, T_v \rangle$.
(5) By Theorem 6.1, $K_{\mathrm{post}} \circ \Pi_{\mathrm{post}}$ is C^7, symplectic, deck-invariant, and injective modulo $\langle T_u, T_v \rangle$ on $\{p < h_K\} \setminus \mathrm{deck}\{\mathcal{M}_1, \mathcal{M}_2, \mathcal{M}_3\}$ into $B^\circ(1)$. Hence $\tilde{f}_i = K_{\mathrm{post}} \circ \Pi_{\mathrm{post}} \circ \Psi \circ F^{(i)} \circ C_i \circ \mathrm{KR}$ are C^4 symplectic embeddings on $B^\circ(a)$ with pairwise disjoint images in $B^\circ(1)$.
(6) Finally, $S_{\mathrm{out}}(y_1, y_2, y_3, y_4) = \sqrt{\pi} R_s (y_1, y_2, y_3, y_4)$ maps $B^\circ(1)$ onto $\mathbb{B}^\circ(R_s) \subset \mathbb{B}^\circ(R_{\mathrm{req}})$ with conformal factor πR_s^2, cancelling Stage 1's $(\pi R_s^2)^{-1}$ so that $f_i^* \omega_0 = \omega_0$.
(7) Part (iii) follows immediately from Lemma 7.4. □

Remark 7.5 (Exact arithmetic versus IEEE-754 `float64`). Theorem 1.1 holds over $\mathbb{R}$ for all $k \geq 10$ and $\varepsilon \in (0, 1)$. In `float64`, `solution(k, eps)` evaluates the analytic margins of Proposition 7.3 on the rounded parameters and raises `ValueError` if machine precision underflows a strict inequality (for instance $\varepsilon \leq 10^{-16}$, where `1.0 - eps == 1.0` makes $a - \rho_b = 0$). Within the `float64`-admissible range, all three scalar Newton solvers run 60 safeguarded iterations on bounded brackets (reaching 53-bit accuracy in ≤ 8 steps once quadratic), and the inverse transfer recovers every moment coordinate without subtraction cancellation. Because gaps of size $O(\varepsilon_c/\sqrt{k})$ sit next to $O(1)$ coordinates, local Lipschitz constants reach 10^8–10^{15} for small ε or large k; arbitrary-precision evaluation via `program/check_embedding_mp.py` verifies the submitted Python source directly to 100+ digits.

8. Explicit Liouville generators, trajectory certificates, and conditioning

Beyond direct map evaluation, a normalised Liouville primitive gives an explicit Hamiltonian-generator formula on the radial–Alexander collar stage and pairs naturally with ODE trajectory certificates. Write $\lambda_0 = \frac{1}{2}(q_1\,dp_1 - p_1\,dq_1 + q_2\,dp_2 - p_2\,dq_2)$, so $d\lambda_0 = \omega_0$ and $\lambda_0(X)[V] = \frac{1}{2}\omega_0(X,V)$. We use $J_0 = \left(\begin{smallmatrix} 0 & I_2 \\ -I_2 & 0 \end{smallmatrix}\right)$ in the coordinate order (q_1,q_2,p_1,p_2), so $X_H = J_0\nabla H$ and $\iota_{X_H}\omega_0 = dH$.

Proposition 8.1 (Liouville generator of radial–Alexander dilation and ODE trajectory certificates)**.** *Let $k \geq 10$, $\varepsilon \in (0,1)$, and $f_0,\dots,f_{k-1}\colon \mathbb{B}^\circ(r_+) \to \mathbb{B}^\circ(R_s)$ be the C^4 symplectic embeddings of Theorem 1.1, with $s_i = (3i,0,0,0)$ $(0 \leq i < k)$.*

(i) (Exact primitive and disjoint radial–Alexander collars.) Put $\alpha_i = f_i^*\lambda_0 - \lambda_0$ and

$$S_i(u) = \int_0^1 \alpha_i(tu)[u]\,dt.$$

This is the unique C^4 primitive with $S_i(0) = 0$ and $dS_i = \alpha_i$. The stage formulas below give local potentials for this primitive; the projective stages require phase lifts with fixed basepoint values and compatible additive constants. For any basepoints $a_i \in \mathbb{B}^\circ(\frac{1}{2})$, any smooth clock $v\colon [t_0,1] \to (0,1]$, and any radii $1 < r_1(v) < r_2(v)$ with $(1-v)a_i + v\mathbb{B}(r_2(v)) \subset \mathbb{B}^\circ(r_+)$, the radial–Alexander dilations $\gamma^{(i)}_{v(t)}(u) = v(t)^{-1}f_i((1-v(t))a_i + v(t)u)$ map the k closed collars $\mathbb{B}(r_2(v(t)))$ onto pairwise disjoint compact sets, with pairwise disjoint open images of their interiors. On open neighbourhoods of $\gamma^{(i)}_{v(t)}(\mathbb{B}(r_1(v(t))))$ they satisfy $\iota_{\partial_t\gamma^{(i)}_{v(t)}}\omega_0 = dG^{(i)}_t$ with exact generator

$$G^{(i)}_t(y) = -\frac{\dot v(t)}{v(t)^3}\Big(2\big(S_i(w) - S_i(a_i)\big) + \omega_0(a_i,w)\Big), \qquad w = f_i^{-1}\big(v(t)y\big).$$

(ii) (Trajectory certificate by ODE uniqueness.) Let $H \in C^2(\mathbb{R}\times\mathbb{R}^4)$ have uniformly compact spatial support, and let $\gamma^{(i)}_t(u)$ $(t \in [0,1],\ u \in \mathbb{B}(1))$ be C^1 curves with $\gamma^{(i)}_0(u) = s_i + u$. Then

$$\phi^t_H(s_i+u) = \gamma^{(i)}_t(u) \qquad \text{for every } t \in [0,1]$$

if and only if

$$\partial_t\gamma^{(i)}_t(u) = J_0\nabla_y H\big(t,\gamma^{(i)}_t(u)\big) \qquad \text{for every } t \in [0,1]. \tag{15}$$

In particular, if $\gamma^{(i)}_1(u) = f_i(u)$, the differential identity (15) certifies $\phi^1_H(s_i+u) = f_i(u)$. More generally, if $X_H(t,\cdot)$ is L-Lipschitz on $\mathbb{R}^4$ for $t \in [0,1]$ and $|\partial_t\gamma^{(i)}_t(u) - X_H(t,\gamma^{(i)}_t(u))| \leq \delta$ for all $t \in [0,1]$, then $|\phi^1_H(s_i+u) - \gamma^{(i)}_1(u)| \leq \delta e^L$; thus a strict target margin $R_{\mathrm{req}} - R_s > 0$ absorbs any rigorously bounded uniform velocity residual $\delta \leq (R_{\mathrm{req}} - R_s)e^{-L}$.

Proof. (i) Since $d\alpha_i = f_i^*\omega_0 - \omega_0 = 0$, the radial Poincaré formula gives $dS_i = \alpha_i$ and $S_i(0) = 0$. The coefficients of α_i are C^3, so this identity gives $S_i \in C^4$; connectedness gives uniqueness. For local stage computations use $\lambda_V = \frac{1}{2}(w_x\,dw_y - w_y\,dw_x) = p\,d\theta$, the cell form $X\,dT + \lambda_V$, the flat form $G\,dv + \lambda_V$, and the post form $\sigma\,d\psi + \lambda_V$. For example the strip primitive relative to these forms is $X(T-\frac{1}{2}) + pC - \frac{1}{2}xy$, with C as in (1); the term $-xy/2$ accounts for the source symmetric radial form. A planar keyhole has potential $P_{KS} = \tau(M(X) - XM'(X))(\Psi(\psi) - \psi) - \frac{1}{2}d(w'_y - w_y)$, where the scalar $d > 0$ is the keyhole's eccentricity parameter. Its four-dimensional suspension adds the conjugate-drift term $-G\tau'(G)g_\tau$; the F_2 potential is $(c(G) - Gc'(G))\vartheta$ relative to the mixed flat form. The twist potentials are bump-integral expressions, and the simplex and ball

charts give piecewise action-angle potentials with a projective phase correction. These local potentials are composed by pullback, including the input/output conformal scale factors. Their additive constants are fixed by the global primitive S_i, rather than by independent principal-argument choices. Line avoidance (Theorem 6.1(ii)) permits local lifts of the projective phase; a lift is unique only after its value at a basepoint is fixed. No global branch-selection evaluator is asserted here. Disjointness of $\gamma_v^{(i)}(\mathbb{B}(r_2(v)))$ follows immediately from Theorem 1.1(ii) by dividing by $v > 0$. Along $\gamma_v^{(i)}(u) = v^{-1} f_i(w)$, with $w = (1-v)a_i + vu$, differentiation gives

$$\partial_v \gamma_v^{(i)}(u) = -v^{-2} f_i(w) + v^{-1} Df_i(w)(u - a_i) = -v^{-2}\big(f_i(w) + Df_i(w)(a_i - w)\big).$$

For any tangent vector $\delta y = v^{-1} Df_i(w)\delta w$ ($w = f_i^{-1}(vy)$), symplecticity $f_i^* \omega_0 = \omega_0$ and $2\, dS_i(w)[\delta w] = \omega_0(f_i(w), Df_i(w)\delta w) - \omega_0(w, \delta w)$ give

$$\begin{aligned}\omega_0\big(\partial_v \gamma_v^{(i)}(u), \delta y\big) &= -v^{-3}\omega_0\big(f_i(w), Df_i(w)\delta w\big) - v^{-3}\omega_0(a_i - w, \delta w) \\ &= -v^{-3}\big(2\, dS_i(w)[\delta w] + \omega_0(a_i, \delta w)\big) = d\Big[-v^{-3}\big(2S_i(w) + \omega_0(a_i, w)\big)\Big](\delta y),\end{aligned}$$

whence $\iota_{\dot v \partial_v \gamma_v^{(i)}} \omega_0 = dG_t^{(i)}$.

(ii) Since $H \in C^2(\mathbb{R} \times \mathbb{R}^4)$ has uniformly compact spatial support, $X_H = J_0 \nabla_y H \in C^1(\mathbb{R} \times \mathbb{R}^4, \mathbb{R}^4)$ is globally Lipschitz in y on $[0,1] \times \mathbb{R}^4$ and vanishes outside a compact ball, so its global flow ϕ_H^t exists uniquely. By definition $\partial_t \phi_H^t(x) = X_H(t, \phi_H^t(x))$ with $\phi_H^0(x) = x$, giving the forward implication, and Picard–Lindelöf uniqueness gives the converse; the quantitative δe^L bound follows from Grönwall's inequality on $\eta(t) = \phi_H^t(s_i + u) - \gamma_t^{(i)}(u)$ ($\eta(0) = 0$, $|\dot\eta| \le \delta + L|\eta|$). (The abstract uniqueness and quantitative certificate statements are also formalised in Lean 4 as `picard_lindelof_curve_unique` and, for C^∞ Hamiltonians, `picard_lindelof_epoch_certificate` in `verification/RK4Convergence.lean`.) □

Remark 8.2 (Scope of global Hamiltonian lifts). Let $\gamma_t^{(i)}$ be a multi-stage symplectic isotopy on source collars connecting $s_i + u$ at $t = 0$ to $\gamma_{v(t)}^{(i)}(u)$ at $t = t_0$ (via rigid translation, polar $\mathrm{Sp}(4)$ paths to $Df_i(a_i)$, and the local Alexander germ $\rho^{-1}(f_i(a_i + \rho(u - a_i)) - f_i(a_i))$ as $\rho \downarrow 0$). Whenever the collar images are pairwise disjoint at every time and each collar uses a consistently normalised single-valued primitive of its exact symplectic generator, multiplying by a plateau cutoff equal to 1 on $\mathbb{B}(r_1)$ and summing over i produces a globally C^3 compactly supported Hamiltonian generating $\gamma_t^{(i)}$. Proving analytic all-(k, ε) clearance bounds for the initial translation/germ collars and single-valued sheet selection in collars near $|Z_0| \to 0$ is left to future work (an experimental prototype and minimal sheet-jump diagnostics are archived outside `program/` in `experiments/hamiltonian_lift/`). Note that Paper I already constructs a globally C^∞ compactly supported packing Hamiltonian for all $k \ge 10$ and $\varepsilon \in (0,1)$, verified in Lean 4.

Remark 8.3 (Numerical conditioning, fixed-step RK4, and variational growth). Along sampled non-toric trajectories (such as $k = 11$, $\varepsilon = 0.05$), the frozen spatial Jacobian $D_y X_H(t,y) = J_0 \nabla_y^2 H(t,y) \in \mathfrak{sp}(4, \mathbb{R})$ exhibits large symmetric eigenvalue pairs $\lambda, -\lambda$ (together with complex conjugates) across narrow keyhole and corner-rounding transitions. These sampled spectra provide local conditioning diagnostics: positive real parts in a frozen Jacobian do not by themselves establish a positive Lyapunov exponent along the time-dependent trajectory, because error amplification is governed by the time-ordered variational equation $\dot\Phi(t) = D_y X_H(t, \gamma(t))\Phi(t) \in \mathrm{Sp}(4, \mathbb{R})$, and these diagnostics do not establish an impossibility result for fixed-precision ODE verification. On the positive side, Paper I, Corollary D (`epoch_rk4_convergence` in `verification/RK4Convergence.lean`) proves that for every $k \ge 10$ and $\varepsilon \in (0,1)$, sufficiently

fine fixed-step RK4 sends every source point into the target ball under controlled initial and per-step state perturbations. Determining the sharp growth rate as $\varepsilon \downarrow 0$ (for fixed non-square $k \geq 10$) of the minimal variational amplification $\sup_{t\in[0,1]} \|D\phi_H^t\|$ over all smooth compactly supported packing Hamiltonians H is an interesting open problem.

Authorship and the role of AI

The manuscripts, mathematical proofs, Lean 4 formalisation and accompanying programs of this release were generated by AI models, including GPT-6.1 Sol (OpenAI), Claude Opus 5.5 (Anthropic), Gemini 4 Carbon (Google) and other models. Liang Song chose the problem, designed and directed the research environment, organised cross-model review and takes sole responsibility for the manuscript and its release.

Trustworthiness rests on explicit verification layers: the main theorems of Papers I–II (along with the abstract trajectory-certificate lemmas of `verification/RK4Convergence.lean`) are checked in Lean 4 against an independent specification importing only Mathlib, using only the three standard axioms and re-verified by fresh kernel replay; Paper III and this paper give human-readable proofs supported by multi-model cross-review and numerical checks (`float64` and 100-digit `mpmath` evaluation documented in `program/README.md`); and program outputs serve only as numerical corroboration.

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